

HYPERVISOR VI

Central Monitoring System

Service Manual

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NOTE

-
- **This manual describes all features and options. The equipment may not have all of them. Contact Mindray service department for any questions.**
-

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2. Freight policy

The customer is responsible for freight charges when this product is shipped to Mindray for service (including any relevant customs fees or other freight related charges).

3. Return address

Please send the part(s) or equipment to the address offered by Customer Service Department.

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FOR YOUR NOTES

1 Safety

1.1 Safety Information

DANGER

- Indicates an imminent hazard situation that, if not avoided, will result in death or serious injury.
-

WARNING

- Indicates a potential hazard situation or unsafe practice that, if not avoided, could result in death or serious injury.
-

CAUTION

- Indicates a potential hazard or unsafe practice that, if not avoided, could result in minor personal injury or product/property damage.
-

NOTE

- Provides application tips or other useful information to ensure that you get the most from your product.
-

1.1.1 Dangers

There are no dangers that refer to the product in general. Specific “Danger” statements may be given in the respective sections of this operation manual

1.1.2 Warnings

WARNING

- **The device is intended for use only by clinical professionals or under their guidance. It must only be used by persons who have received adequate training in its use. Anyone unauthorized or untrained must not perform any operation on it.**
 - **The CMS is a clinical information device. Except for using such components as the mouse and keyboard to perform normal operations, do not touch or disassemble any other component, especially the power component; otherwise, it may result in personnel injury.**
 - **Do not connect this system to outlets with the same circuit breakers and fuses that control current to devices such as life-support systems. If this system malfunctions and generates an overcurrent, or when there is an instantaneous current at power ON, the circuit breakers and fuses of the building’s supply circuit may be tripped.**
 - **Failure on the part of the responsible hospital or institution employing the use of the CMS to implement a satisfactory maintenance schedule may cause undue equipment failure and possible health hazard.**
 - **Be sure to keep the packaging materials from children’s reach. Disposal of the packaging materials shall comply with your local requirements.**
 - **If any value displayed on the screen of the CMS is abnormal or questionable, first determine the patient’s vital signs by alternative means and then verify that the CMS or monitor is working correctly.**
 - **The physiological waveforms, parameters and alarms displayed on the CMS are transmitted from the monitor through the network. If there is a network failure, the data loss or delay may occur. Pay close attention to the patients during a network failure.**
-

1.1.3 Cautions

CAUTION

- Hospitals without stable power source should use an Uninterruptible Power Supply (UPS) to power the CMS. When there is a power failure, the system should be shut down by following the specified shutdown procedure before the UPS is turned off. If the system has a sudden power failure, system failure may occur and consequently the system will not work correctly next time or even have a serious result.
 - The host of the CMS should be installed with the original Microsoft Windows's system and standard upgrade program, such as the service package. Illegal software may lead to abnormal or incorrect system operating.
 - The host of the CMS should be maintained every three to six months. Its long time continuous operating may lead to failure of the operating system.
 - Protect the device from damage caused by drop, impact, strong vibration or other mechanical force during servicing.
-

1.1.4 Notes

NOTE

- Refer to the Operator's manual for more information.
-

1.2 Equipment Symbols

	Refer to instruction manual/booklet.
	CAUTION: To reduce the risk of electric shock, do NOT remove. cover. Refer servicing to qualified service personnel.
	Alternating current (AC)
	Power switch
	Keyboard port

	Mouse port
	Serial communication (COM) port
	Display port
	Printer port
	USB port or device
	Network port
	Sound output port
	Sound input port
	Microphone port
	Manufacture date
	MANUFACTURER
	Serial number

	Equipotentiality
	The product bears CE mark indicating its conformity with the provisions of the Council Directive 93/42/EEC concerning medical devices and fulfils the essential requirements of Annex I of this directive. 0123 is the number of the EU-notified body. Note: The product complies with the Council Directive 2011/65/EU.
	The following definition of the WEEE label applies to EU member states only. This symbol indicates that this product should not be treated as household waste. By ensuring that this product is disposed of correctly, you will help prevent bringing potential negative consequences to the environment and human health. For more detailed information with regard to returning and recycling this product, please consult the distributor from whom you purchased it. * For system products, this label may be attached to the main unit only.

FOR YOUR NOTES

2 Introduction

2.1 Overview

HYPERVISOR VI Central Monitoring System (hereinafter called CMS) is intended for professional physicians or paramedics to conduct centralized monitoring over patients monitored by Mindray individual monitors and/or telemetry systems in hospitals or medical institutions.

The central monitoring system comprises powerful system software and high-performance computer. It constructs a monitoring network by connecting monitors and/or telemetry. By collecting, processing, analyzing and outputting the information coming from monitors and/or telemetry, the central monitoring system can achieve centralized monitoring over multiple patients so as to greatly promote the efficiency and quality of the monitoring work.

2.2 CMS Recommended Configuration

CMS Hardware components should be highly reliable and stable. Refer to corresponding Operator's Manual for recommended hardware configuration.

2.3 Intellectual Property Protection

The HYPERVISOR VI Central Monitoring System uses a USB dongle for intellectual property protection. You must plug the dongle into the system's USB interface before starting the system. Otherwise, the system cannot start.

NOTE

- **Before you use the USB dongle, please install driver first.**
 - **When installing or using the CMS, you must plug in the USB dongle.**
-

If the dongle is damaged or lost, you may need to reinstall the system software before using a new one.

NOTE

- **When reinstalling the system software, try not to remove the old database so as to keep the old monitoring data.**
-

FOR YOUR NOTES

3 System Installation

3.1 Pre-installation Preparations

3.1.1 Environmental Requirements

The CMS should be installed in an environment where the system can be easily viewed, operated and maintained.

The environment where the CMS is installed should be reasonably free from noises, vibration, dust, corrosive, flammable and explosive substances.

If the CMS is installed in a cabinet, sufficient space in front and behind should be left for convenient operation, maintenance and repair. Moreover, to maintain good ventilation, the CMS should be at least 2 inches (5cm) away from around the cabinet.

When the CMS is moved from one place to another, condensation may occur as a result of temperature or humidity difference. In this case, never start the system before the condensation disappears.

3.1.2 Power Requirements

Each component of the CMS must be powered by the specified power source.

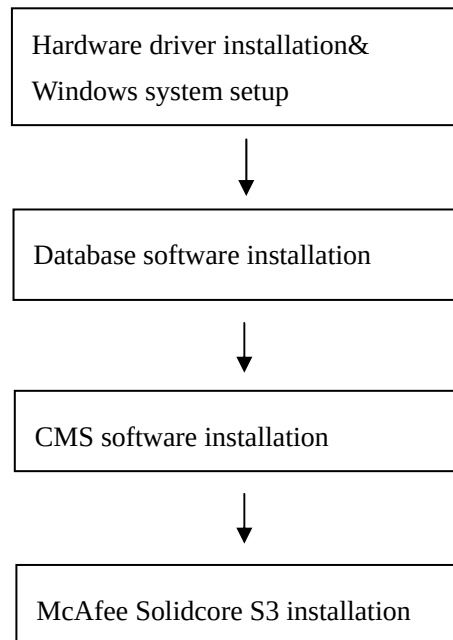
To protect the hospital personnel from electric shock, the CMS (including the host and displays) and its recorder must have their casings properly grounded. The host of the CMS is provided with a 3-wire power cable, which must be plugged into a properly grounded 3-wire receptacle. If a 3-wire, grounded receptacle is not available, consult the hospital electrician.

WARNING

- **Make sure that the operating environment and power source of the CMS meet the specific requirements; otherwise, unexpected consequences, e.g. damage to the equipment, may result.**
 - **Appropriate power supply must be selected according to the setup of the system power voltage; otherwise, serious damage may be caused to the system.**
 - **Never use a 3-wire to 2-wire adapter with any unit of the CMS.**
-

3.2 Installation Procedures

CMS installation procedures include hardware driver installation & Windows system setup, database software MySQL installation, CMS software installation, etc, as shown below.



WARNING

- The CMS host can not be installed with any other software besides the Windows system, necessary drivers, and drivers/software listed in this manual. Otherwise, normal operation of CMS may be affected and unexpected consequences may result.

NOTE

- The CMS software only supports Windows® 7 Professional Embedded SP1 (32 bit) operation system.
- Before performing the operations described below, make sure that the main unit is not installed with any application software except the accompanying software of Windows.
- When the CMS software is installed on the virtual machine of hospital, make sure that the allocated hard disk resource and virtual machine environment meet the CMS configuration requirement. Please contact service representative if you have any questions.

3.3 Installation Preparations

3.3.1 CD Preparation

The following software is needed for the installation of CMS software:

- Windows 7 Pro Embedded SP1 (32 bit) Operating System software CD (G-110-003463-00)
- Operation System Installation CD patches
- CMS software CD (G-110-003916-00)

3.3.2 BIOS Configuration

1. Set BIOS password to “Hypernet” (case sensitive).
2. Set the storage mode:
 - ◆ For single hard disk computer, set “SATA Emulation” to “AHCI”.
 - ◆ For dual hard disk computer, set “SATA Emulation” to “RAID”.
3. Set start mode to starting from hard disk first.

3.3.3 System Fault Recovery

For errors occurring during installation of application software and setup, quick recovery method can be used to recover the system. For details, refer to **Chapter 6 System Recovery**. In case of recovery failure, contact the computer supplier. Proceed with the following operations if the system returns normal.

3.4 Hard Disk Partition

The hard disk must have three (3) partitions:

- Disk C: greater than 60 G
- Disk D: greater than 380 G
- Disk E: greater than 60 G

The file systems of disks C and D must be in the NTFS format and that of disk E in the FAT32 format.

NOTE

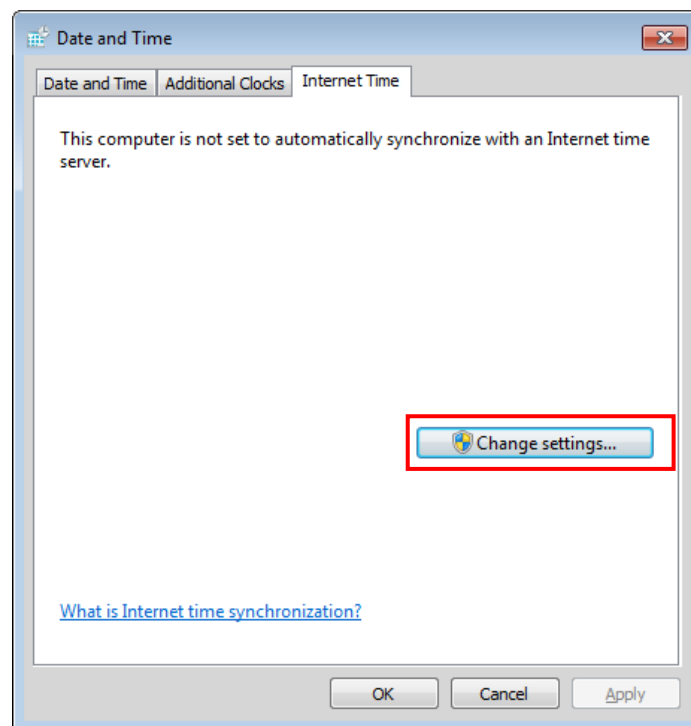
-
- **The file system of the disk partition where the data of the database is located must be in the NTFS format.**
-

3.5 Installing Operating System

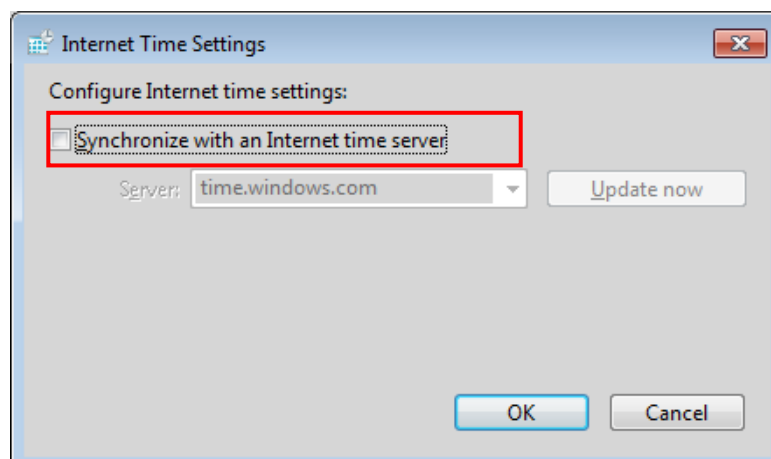
Refer to the Windows installation instructions for details.

3.6 Time Configuration in Operating System

1. Click the time of operating system at the lower right corner of screen
2. Select “Change date and time settings”. Then “Date and Time” window will pop up.
3. Select “Internet Time” tab, and then select “Change settings...”, as shown below.



4. Please ensure that the “Synchronize with an Internet time server” is not selected in “Internet Time” tab, shown as below.



3.7 Display Hardware Installation (Optional)

3.7.1 Double Screen Display Adapter



-
- **Disconnect the power cord of the host before installing the PCI double screen display adapter.**
-

If it is necessary to install a double screen display adapter, do the following:

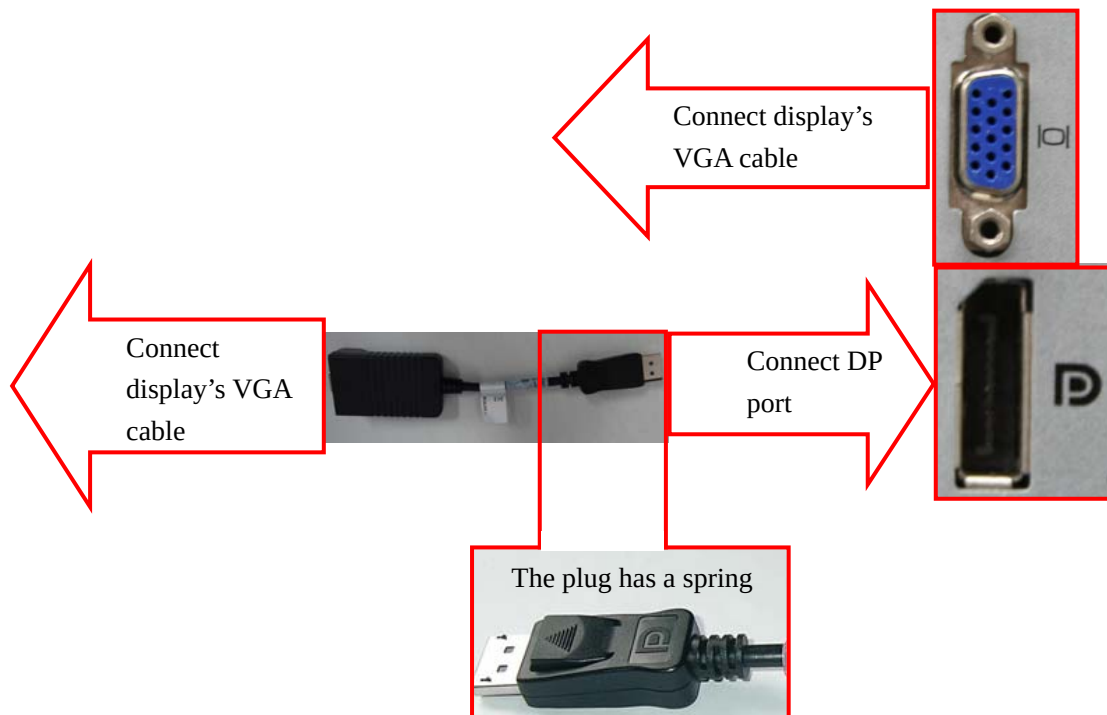
1. Install the double screen display adapter: Disconnect the power supply; insert the double screen display adapter into the PCI slot on the main board and then connect the primary and secondary display respectively to port 1 and port 2 of the double screen display adapter respectively.
2. Install display adapter driver: Refer to the accompanying documents of the display adapter for details. Restart the computer after the display adapter driver has been installed.
3. Set double screen display mode: Refer to the accompanying documents of the display adapter for details.

3.7.2 Implement Double Screen Display by DP-to-VGA

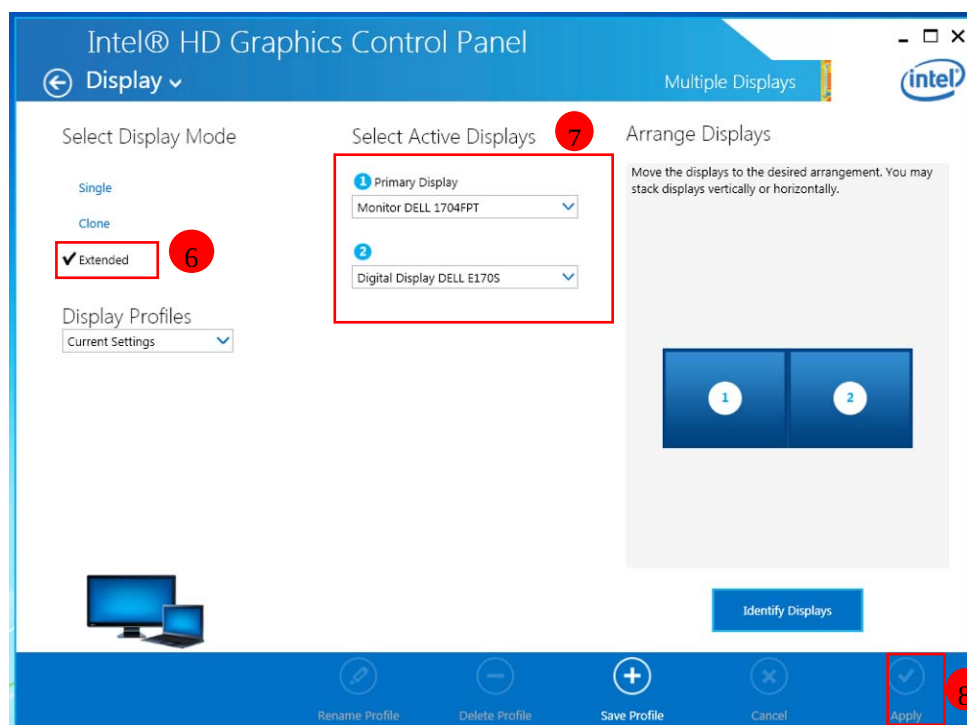
Adapter

The double screen display in the CMS can be achieved by DP-to-VGA adapter:

1. Shut down the computer first.
2. Then connect two screens by DP-to-VGA adapter.



3. When the display is connected, power on the PC.
4. Right click the mouse on the desktop, and then select “Graphics Properties...” from the pop-up menu.
5. In the “Intel® HD Graphics Control Panel” window, click “Display” → “Multiple Displays” to enter the “Multiple Displays” window, as shown in the following figure.



6. In the “Select Display Mode” field, select “Extended”.

7. In the “Select Active Displays” field, set the primary display to “Monitor ***”; set the second display to “Digital Display***”.
“Monitor ***” refers to the display directly connecting the VGA interface of the PC.
“Digital Display***” refers to the display connecting the VGA by DP-to-VGA adapter.
8. Click the “Apply” option. The “The new settings have been applied. Do you want to keep these settings?” message displays.
9. Click the “Yes” button to apply the setting.
10. Click the  icon to close the window.

NOTE

- When unplugging the DP-to-VGA adapter, press the spring button on the DP plug and then unplug it downward. Failure to do so may damage the DP plug.

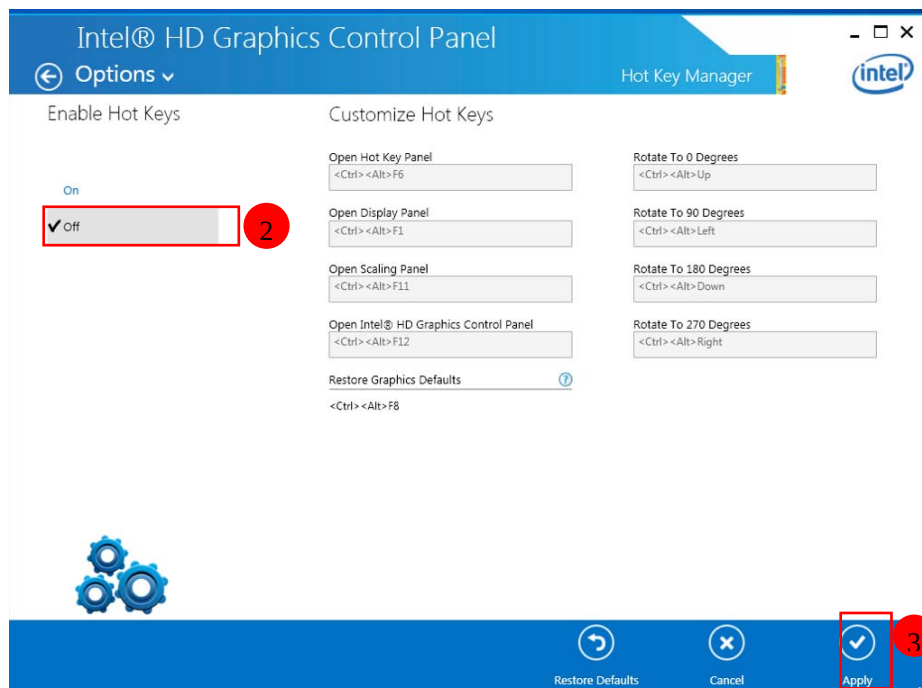


The label beside DP port is:

3.7.3 Cancelling the Hot Keys

To avoid the user error operation, follow the procedure to cancel the functional hot keys:

1. Enter the “Intel ® HD Graphics Control Panel” window, and then click the “Options” option to display the “Hot Key Manager” window, as shown in the following figure.

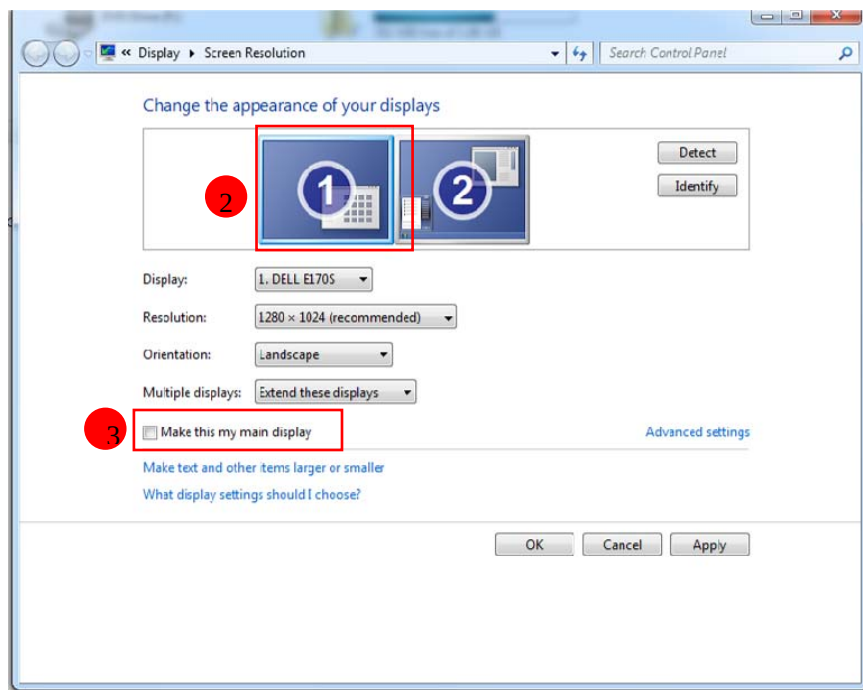


2. In the “Enable Hot Keys” field, click the “Off” option.
3. Click the “Apply” option to apply the setting. The “The new settings have been applied. Do you want to keep these settings?” message displays.
4. Click the “Yes” button to apply the setting.
5. Click the ✕ icon to close the window.

3.7.4 Setting My Main Display

To make the desired screen as main display, follow this procedure:

1. Right click the mouse on the desktop, and then select “Screen resolution” from the pop-up menu to display the window, as shown in the following figure.

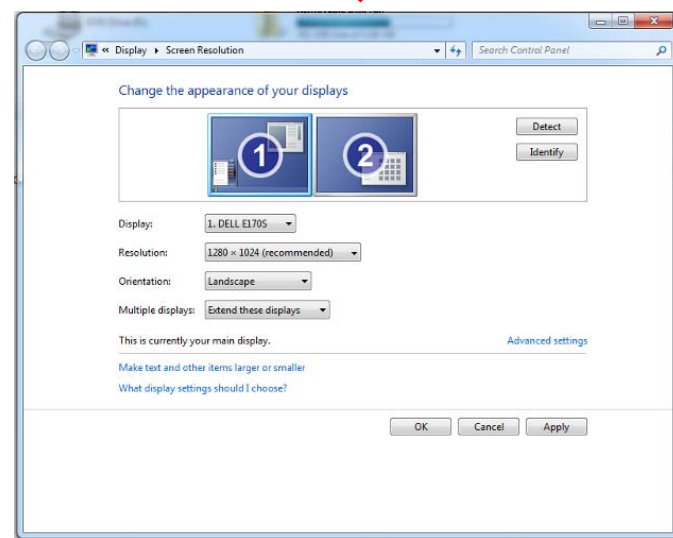
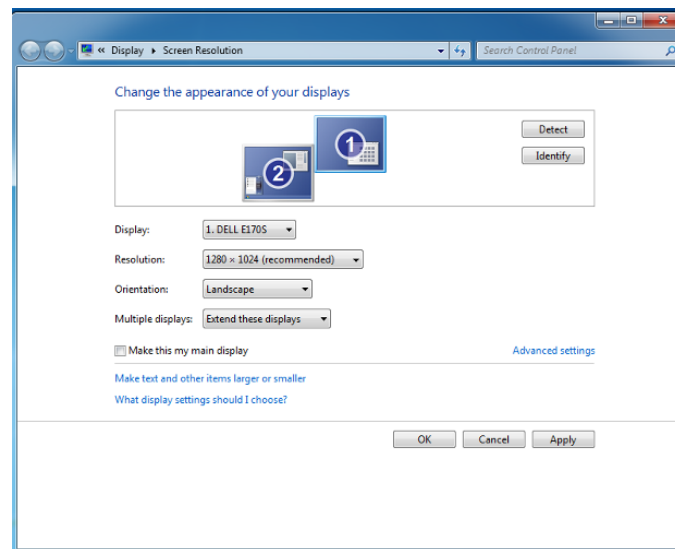


2. In the screens displaying list box, select the desired screen.
3. Select the “Make this my main display” option.
4. Click the “OK” button to save the setting and close the window.

3.7.5 Aligning the Double Screens

When the alignment between the screens is ragged, follow this procedure to align the screens at the same horizontal line:

1. Right click the mouse on the desktop, and then select “Screen resolution” from the pop-up menu to display the window, as shown in the following figure.

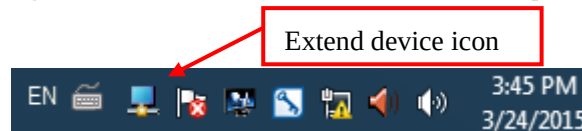


2. Click the screen signed with number 1 to select it, and then drag the screen down until the screen signed with number 1 aligns with the screen signed with number 2.
3. Click the “OK” button to save the setting and close the window.

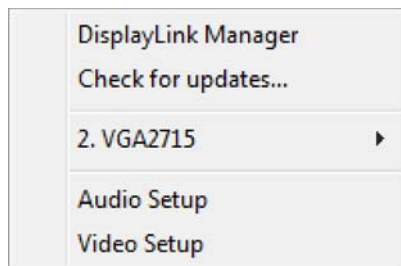
You can refer the above steps to adjust the multi-screens alignment.

3.7.6 Installing Multi-Screen Extend Device (SUNIX VGA2715)

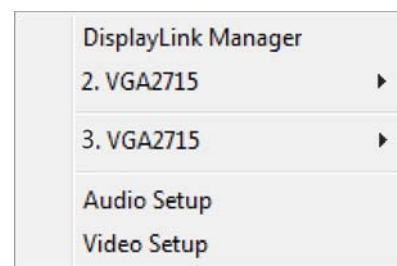
1. Take out the accompanying USB cable. Connect one end of the cable with the connector of SUNIX VGA2715, and the other end with the CMS.
2. Place the SUNIX VGA2715 driver CD in the CMS drive. Double click the driver in the CD and install by default configuration. The following icon will display in menu bar on the lower right corner of screen when installation is complete.



3. Connect the VGA cable of display with SUNIX VGA2715. Right click VGA2715 icon in the menu bar. When the CMS are equipped with three or four displays, the following menu will pop up.

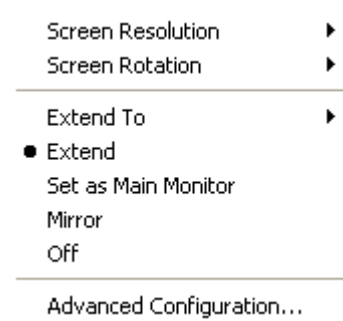


Three display menu

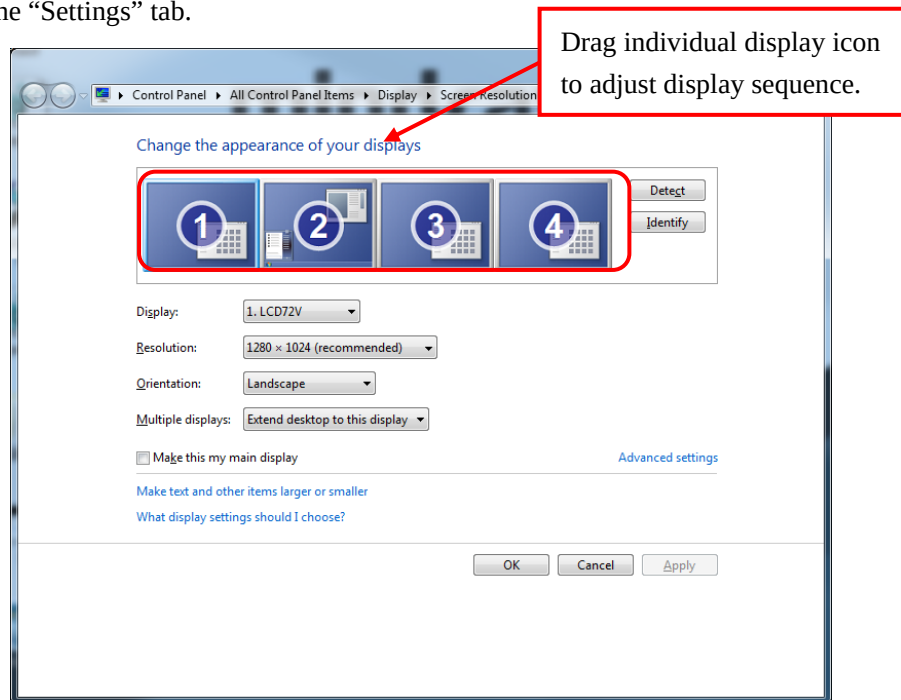


Four display menu

4. Select the arrow to open the submenu.
5. Select the option of “Extend” in the corresponding submenu. Then the extension of the screen is done.



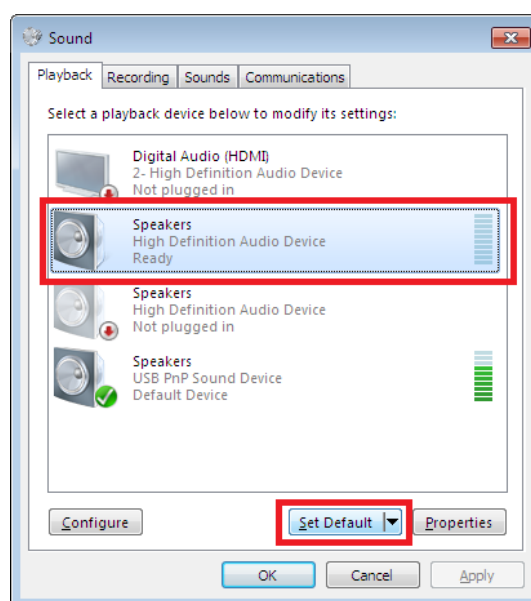
- To adjust the display sequence, select the option of Advanced from the pop-up menu or sub-menu. The following Display Properties window will show. Set the sequence in the “Settings” tab.



3.7.7 Setting the Audio Properties (T100 Extend Device Only)

When the T100 display extend device is used, you need to set the audio.

- Open “Control Panel”, and then select “Sound”. The “Sound” window will display.
- In the “Playback” tab, select the operating system speaker and then select “Set Default” button, shown as below.



-
3. Restart PC when the above installation is done. Verify that the display effect meets the requirement.

3.7.8 Installation for Remote Display

The CMS supports the display of local video on remote display through VGA video extension device.

Transmitter

1. Connect the CMS host with the INPUTS connector of the transmitter by a VGA cable.
2. Connect the local display of the CMS with OUTPUTS connector of transmitter by a VGA cable.
3. Connect one end of network cable with the RJ45 connector of transmitter.
4. Connect the power adapter with the POWER connector of transmitter.

Receiver

1. Connect the input port of remote equipment (such as a display or a LCD TV) with the Output connector of the receiver by a VGA cable.
2. Connect one end of network cable with the RJ45 connector of receiver.
3. Connect the power adapter with the POWER connector of receiver.

After installation, you can finely adjust the brightness and contrast through “Brightness” and “Contrast” buttons on the receiver.

NOTE

-
- **The CMS supports VGA video extension. The recommended equipment is ATEN KVM Extender CE750.**
 - **If multiplex video signals are required to be displayed in separate screens, use other types of ATEN transmission equipment.**
 - **The remote display equipment is required to support a resolution of 1280×1024 pixels.**
 - **Please refer to the accompanying documents for the detailed operations, installation, specifications and precautions of VGA video transmission equipment.**
-

3.7.9 Installation for Remote USB Console

The CMS can achieve the control for a remote USB device through USB twisted pair cable transmission device.

1. Firmly connect the USB-B port of ATEN CE750L unit with the USB port of the CMS through a USB 2.0 wire.
2. Connect the USB devices (such as keyboard, mouse) with the ATEN CE750R unit.
3. Connect ATEN CE750L with ATEN CE750R by a network cable.
4. Verify that the Remote LED lamp (green) of ATEN CE750R unit is On.

NOTE

- **The CMS supports the transmission from a remote USB console. The recommended equipment is ATEN KVM Extender CE750.**
 - **When the remote USB device is installed, it needs 1 to 2 minutes to implement USB self-test. Please wait for a while.**
 - **Please refer to the ATEN CE750 accompanying documents for the detailed operations, installation, specifications and precautions.**
-

3.8 IP Address Setup and Network Connection

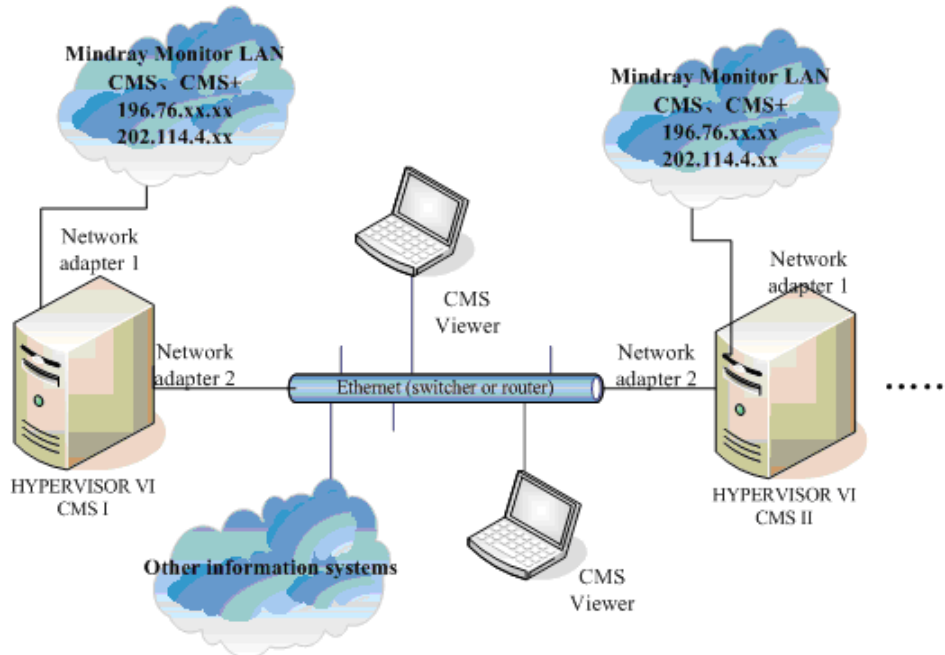
The PC may be equipped with two network adapters to isolate the monitor LAN and the external network to ensure network bandwidth and data safety for the monitors.

NOTE

- **If two network adapters are used, be sure to identify which is for monitor network and which is for external network connecting multiple CMSs, CMS viewer or other information systems. Correctly set IP address for each network adapter.**
 - **If two network adapters are used, connect them to corresponding networks as desired.**
 - **Do not connect both network adapters to the same network segment, e.g., do not connect them to the monitor network at same time.**
-

3.8.1 Network Connection using Multiple Network adapters

The figure below shows the network connection using multiple network adapters.



As shown in the figure, the two network adapters are respectively connected to the monitor LAN and the external network.

3.8.2 Monitor LAN

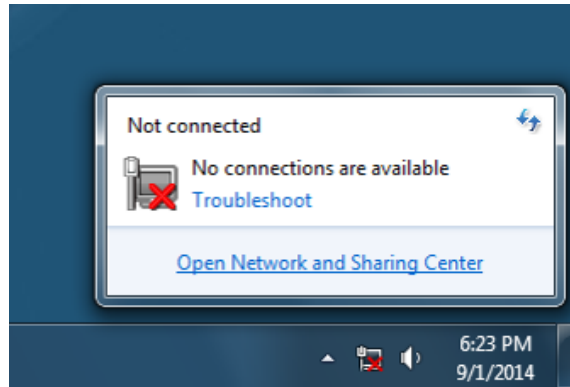
NOTE

- If two network adapters are used, the one (hereinafter called integrated network adapter) integrated on the PC main board is normally connected to the monitor network and the one (hereinafter called independent network adapter) installed in the PCI slot is connected to the external network.
 - For the CMS protocol, the IP Address 202.114.4.119 must be configured.
 - Every IP Address on each individual network has to be unique. Duplicate IP Addresses will cause network connection failures .
 - The hospitals can set their own desired DHCP or IP addresses. However, the CMS IP address must not be changed.
-

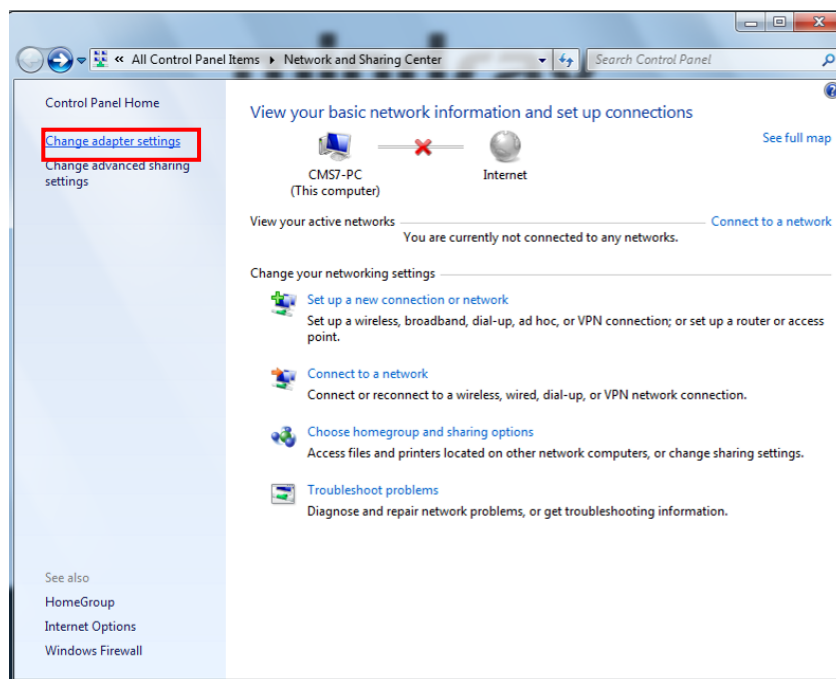
To reduce network bandwidth consumption, IP address should be configured according to the monitor and protocol type on the principle that fewer IP addresses, maximum 2, are configured for each network adapter.

To set the IP addresses of the integrated network adapter:

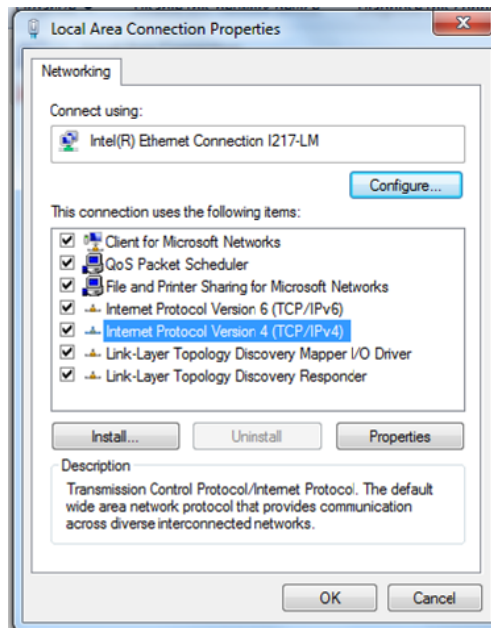
1. Select the  icon at the bottom of the screen to enter the following screen. Then click “Open Network and Sharing Center”.



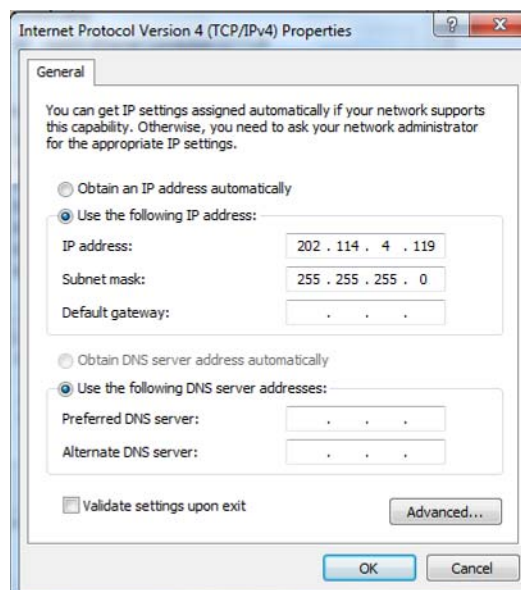
2. In the pop-up screen, select “Change adapter settings”.



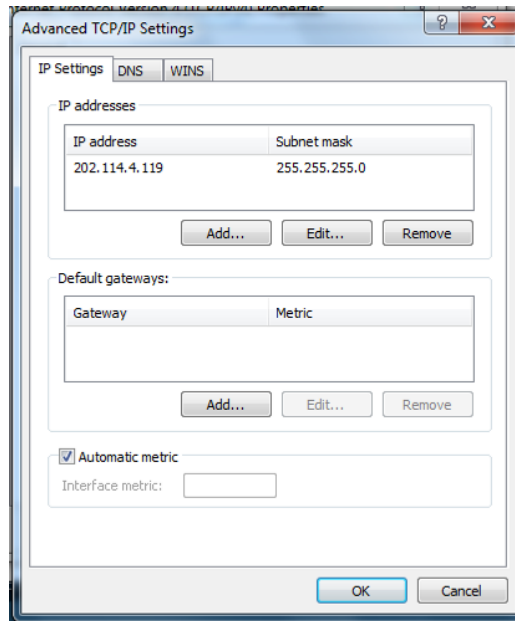
3. After accessing the Network Connection screen, right click “Local Area Connection”.
4. In the “Local Area Connection Properties” screen, double click “Internet Protocol version 4 (TCP/IPv4)”.



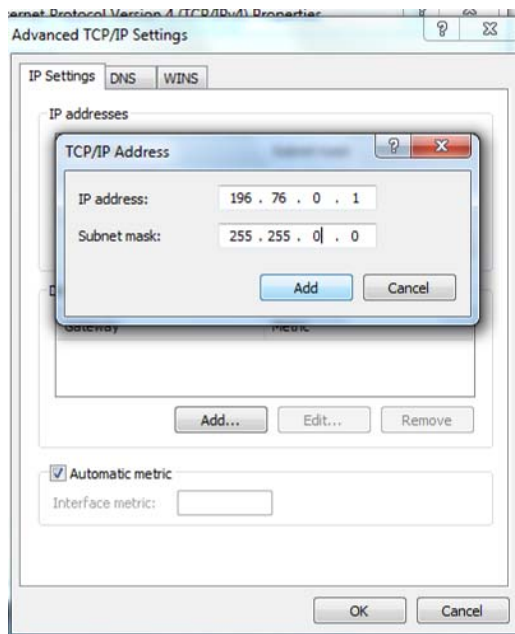
5. In the “Internet Protocol Version 4 (TCP/IPv4) Properties” screen, set an IP address (The address in the following figure is used for reference only).



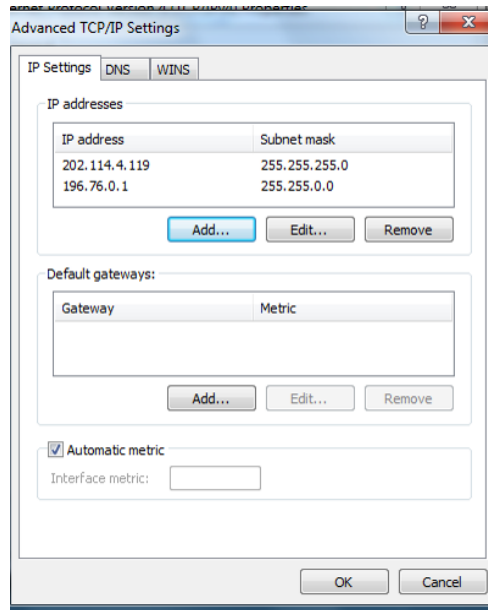
6. If you need to configure more IP addresses, select “Advanced” (see the figure above), you will see the following screen display:



7. Click “Add...” in the “IP Settings”, and then you will see the following screen display:



8. Click “Add” to return to the following screen display. To add more IP addresses, repeat Step 7.



9. Click “OK” to finish setting IP address.

3.8.3 External Network

The independent network adapter is used to connect the external network. Its IP address should be configured according to actual use.

The external network is required when:

- Other information systems are connected;
- Patients of different CMSs are viewed;
- Local CMS is remotely viewed using a CMS viewer;
- Routers are used.

IP Address Setup

To implement the Remote CMS function between CMSs, each CMS needs to be equipped with at least two network adapters, one of which is for internal communication within the current CMS and the other for inter-communication between the CMSs. The typical configuration is shown below:

Current CMS A		Target CMS B		Target CMS C	
Network adapter A1	Network adapter A2	Network adapter B1	Network adapter B2	Network adapter C1	Network adapter C2
196.76.0.X1	168.168.4.X2	196.76.0.Y1	168.168.4.Y2	196.76.0.Z1	168.168.4.Z2

As shown in the table above, network adapter A1, B1 and C1 are used for the communication between the CMS and the monitors. Their IP addresses are in the same network segment with the IP addresses of respective monitors. Network adapter A2, B2 and

C2 are used for the inter-communication between the CMSs. Their IP addresses belong to the same network segment and X2, Y2 and Z2 shall be different.

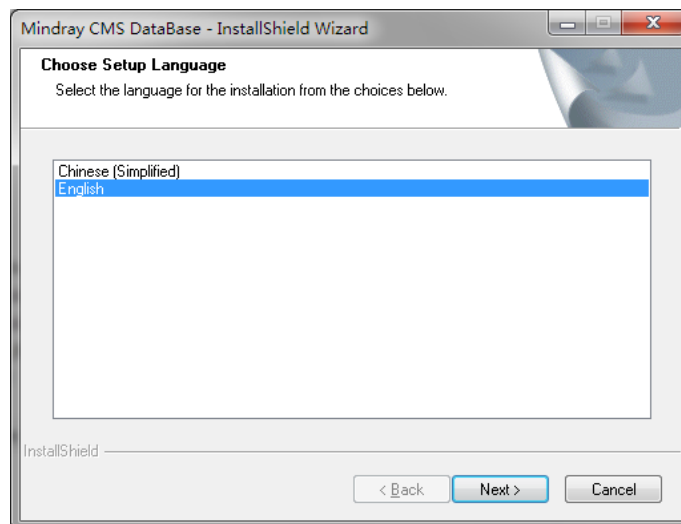
NOTE

- **If a CMS needs to implement the Remote CMS function on other CMS, other CMS has to connect to the monitoring network. Please refer to CMS operator's manual for details.**
-

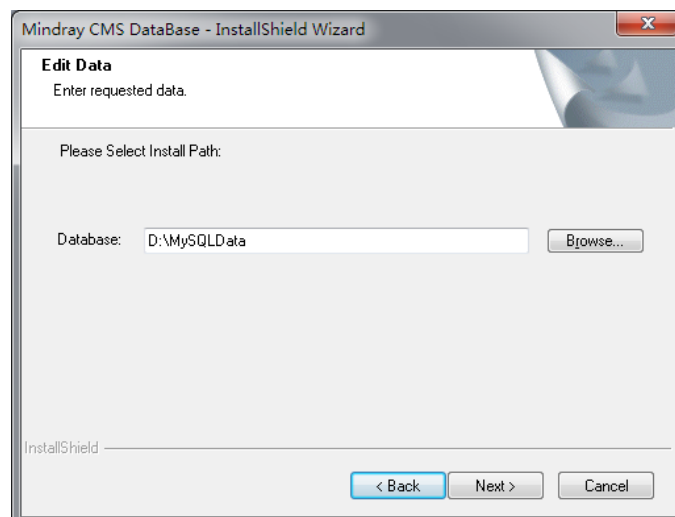
3.9 Installing Database Software

Insert the CMS software CD, then go to the directory “MySQL Community Server” and install by following the steps. Restart the PC after each step is completed.

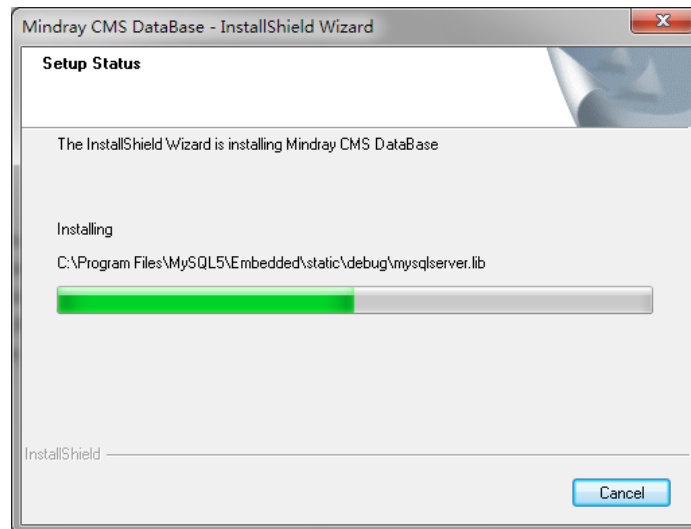
1. Double click MySQLInstall.exe, and the following “Choose Setup Language” screen will be displayed:



2. Select a desired language and click “Next” button till the following screen is shown:

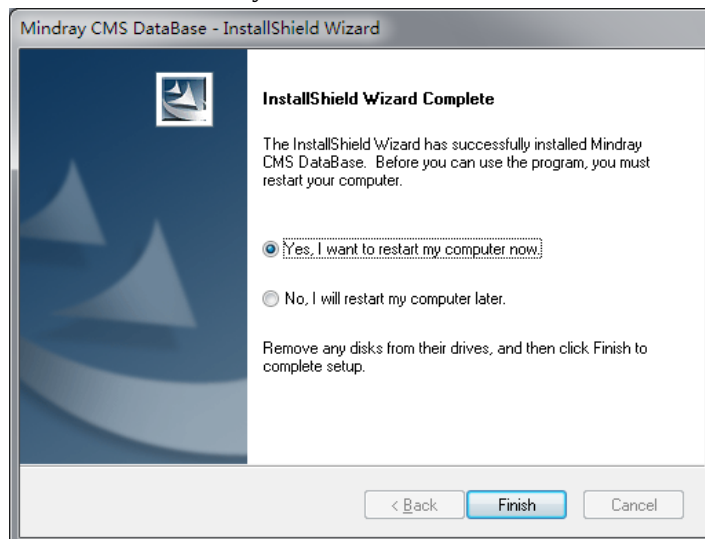


Since the data of database is saved under this directory, make sure that the disk in which the directory is located can read-write and provide enough space. When the selection is completed, click “Next” button until the following screen is shown:



3. Complete installation and restart the PC

When the following screen is shown, it indicates that the database software installation is complete. Select as the following screen shows and then click the “Finish” button. The computer will restart automatically.



3.10 Installing Printer (Optional)

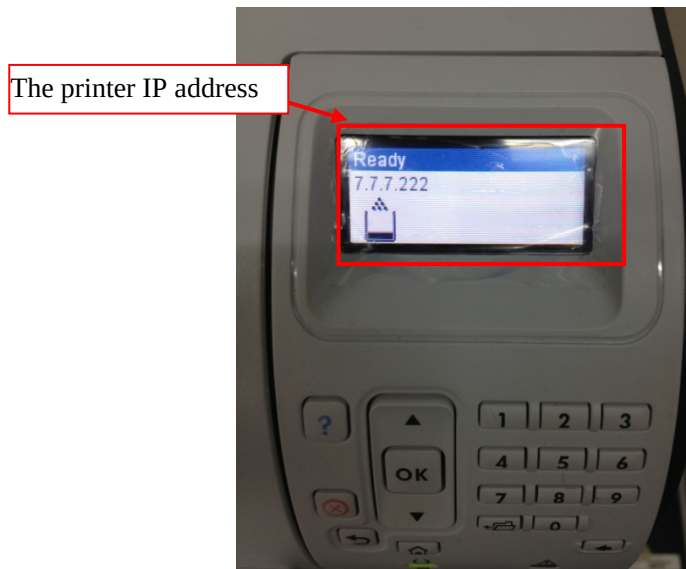
3.10.1 Printer Installation Procedures

Take HP LaserJet M602 as an example.

Insert the printer driver CD into the CD-ROM.

Verifying the Printer IP:

1. Load the printer with paper and connect with the computer (connect to the integrated network card with IP address 196.76.0.X).
2. Power on the printer. The printer IP address information will be shown on the screen of the control panel as below:



3. Verify this IP address is in the same network segment with the computer. If not, add a new computer TCP/IP address so that it is in the same network segment with the printer IP address.

Adding TCP/IP

1. Enter the Windows desktop and select “Start” → “Control Panel” → “Network and Internet” → “Network and Sharing Center” → “Change adapter settings” to display the network connection window.
2. Double-click the “Local Area Connection” icon to display the “Local Area Connection Status” window.
3. Click the “Properties” button to display the “Local Area Connection Properties” window.

-
4. Select the desired device from the “This connection uses the following items” list box, and then click the “Properties” button to display the “Internet Protocol Version 4 (TCP/IPv4) Properties” window.
 5. Select the “Use the following IPv4 address” option, and then click the “Advanced...” button to display the “Advanced TCP/IP Setting” window.
 6. Select the “IP Settings” tab, and then click the “Add...” button to display the “TCP/IP Address” window.
 7. Input an IP address that is in the same network segment with the printer IP address. Click the “Subnet mask” input box, the default subnet mask displays.
 8. Click the “Add” button to add the IP and close the “TCP/IP Address” window.
 9. Click the “OK” button to save the setting and close the “Advanced TCP/IP Setting” window.
 10. Click the “OK” button to save the setting and close the “Internet Protocol Version 4 (TCP/IPv4) Properties” window.
 11. Click the “OK” button to apply the setting and close the “Local Area Connection Properties” window.
 12. Click the “Close” button to close the “Local Area Connection Status” window.

Adding or Editing Additional Applications

If applications have to be added, deleted or updated after the installation of Solidcore S3 Control, perform the following to enable the changes under the control of Solidcore:

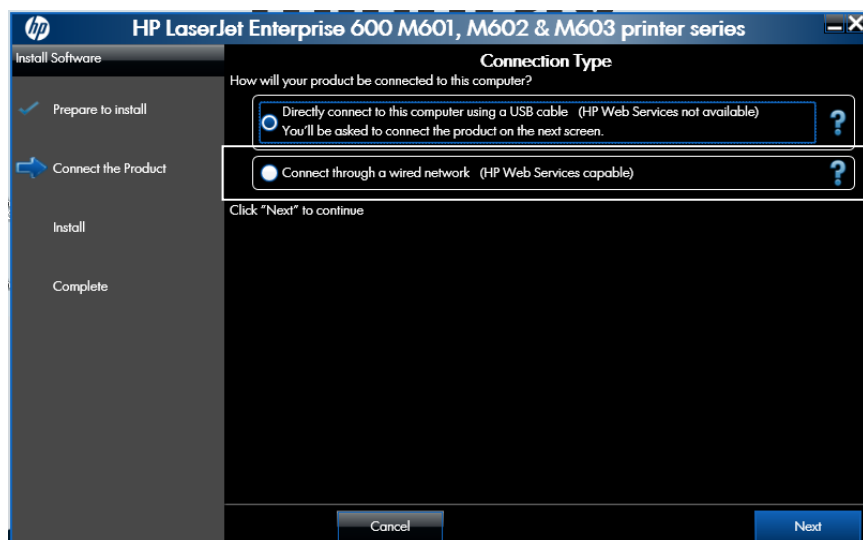
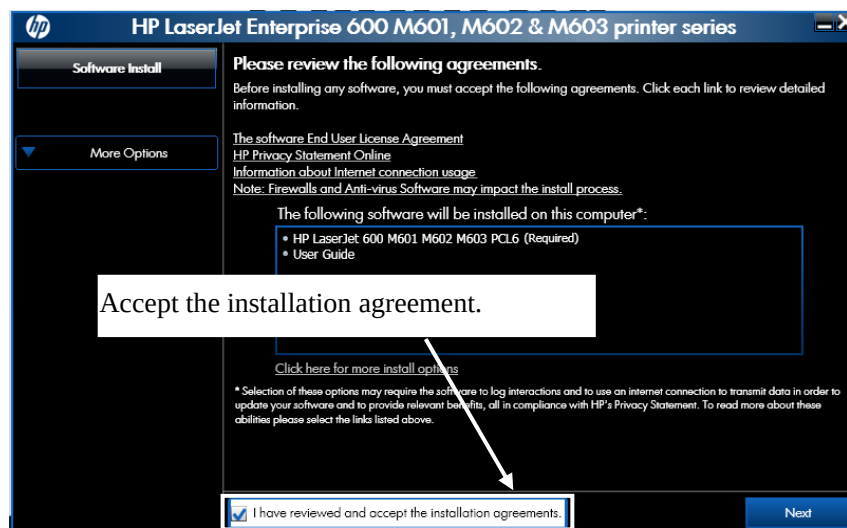
1. Double click the icon of McAfee Solidifier Command Line on the desktop and the corresponding window will display.
2. Input "sadmin disable" in the command line, press the “Enter” key, and then restart the PC.
The Solidcore control is disabled. You can add or delete the additional applications.
3. Double click the icon of McAfee Solidifier Command Line on the desktop, and then input "sadmin so" to scan all of applications.
4. Input “sadmin enable” in the command line, press the “Enter” key, and then restart the PC.

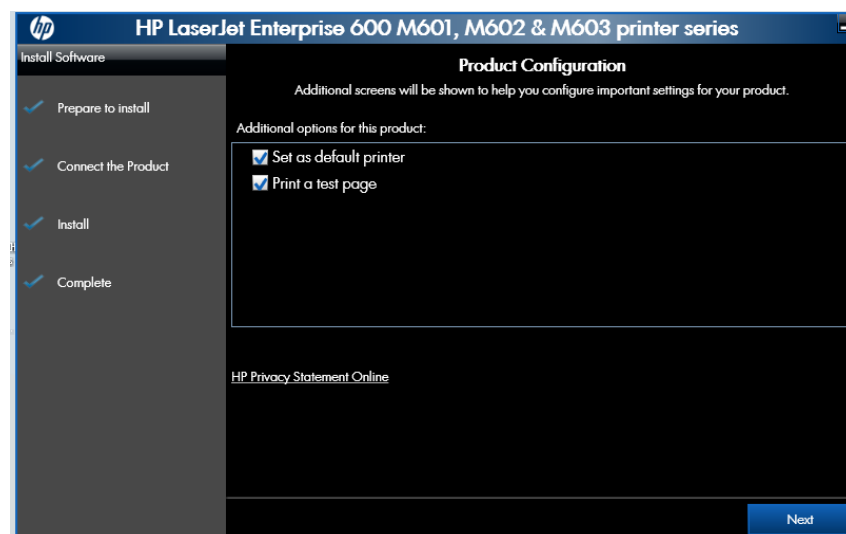
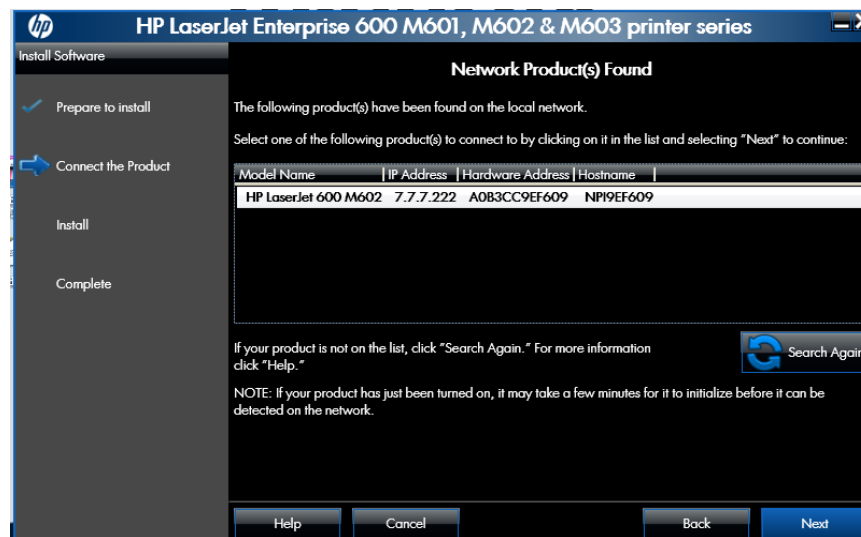
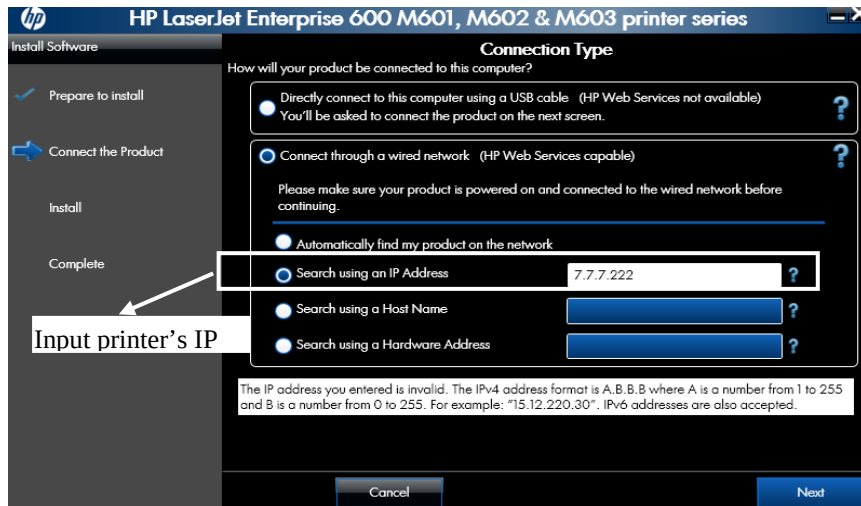
Other commonly used commands of McAfee Solidifier include:

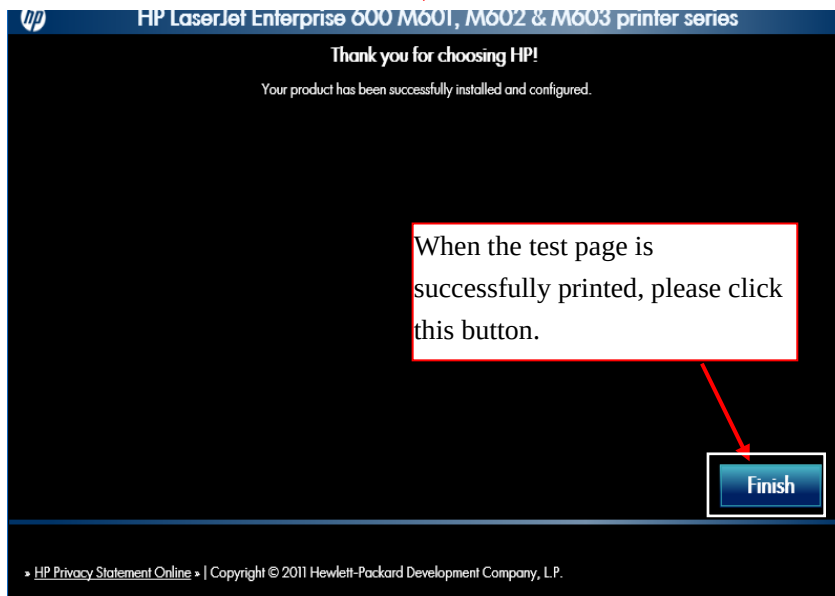
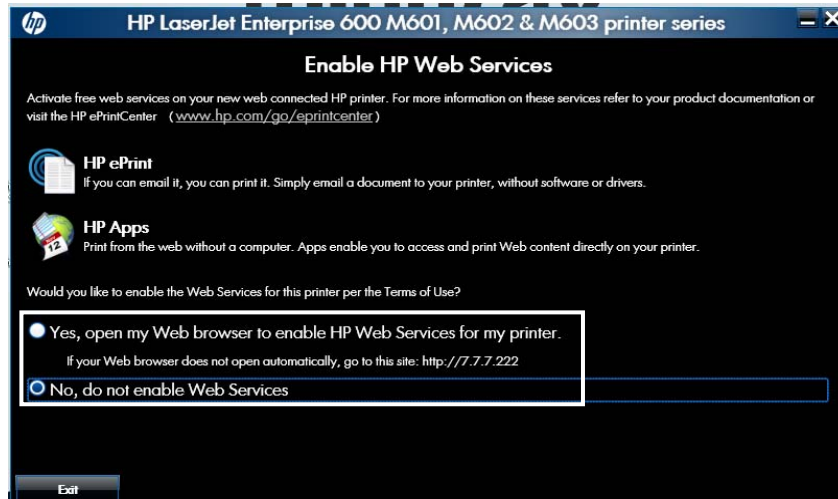
- sadmin help: used to view the commonly used commands;
- sadmin status: used to view the status of McAfee.

Installing the Network Printer

1. Insert the printer driver CD into the CD-ROM.
2. Right click the mouse to run CD driver, and then double click “setup.exe” file in the root directory.
3. Install the driver by following the indications shown below in the order of from left to right and from top to bottom:







4. After test page is printed, close all the pop-up windows and access “Printers and Faxes”.
5. Select “HP LaserJet 600 M601 M602 M603 PCL6” printer and re-name it as “M602_net” to complete installation of printer’s network driver.

Adding or Editing Additional Applications

If applications have to be added, deleted or updated after the installation of Solidcore S3 Control, perform the following to enable the changes under the control of Solidcore:

1. Double click the icon of McAfee Solidifier Command Line on the desktop and the corresponding window will display.
2. Input "sadmin disable" in the command line, press the "Enter" key, and then restart the PC.
The Solidcore control is disabled. You can add or delete the additional applications.
3. Double click the icon of McAfee Solidifier Command Line on the desktop, and then input "sadmin so" to scan all of applications.
4. Input "sadmin enable" in the command line, press the "Enter" key, and then restart the PC.

Other commonly used commands of McAfee Solidifier include:

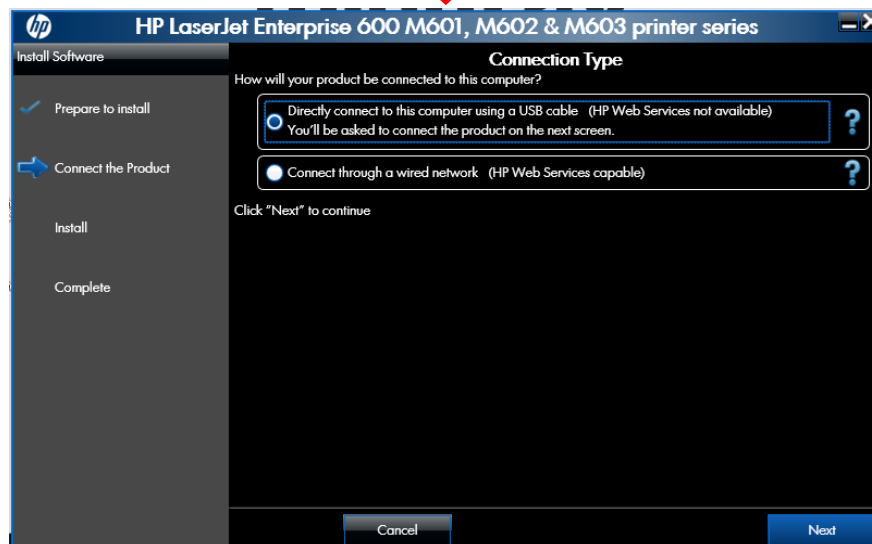
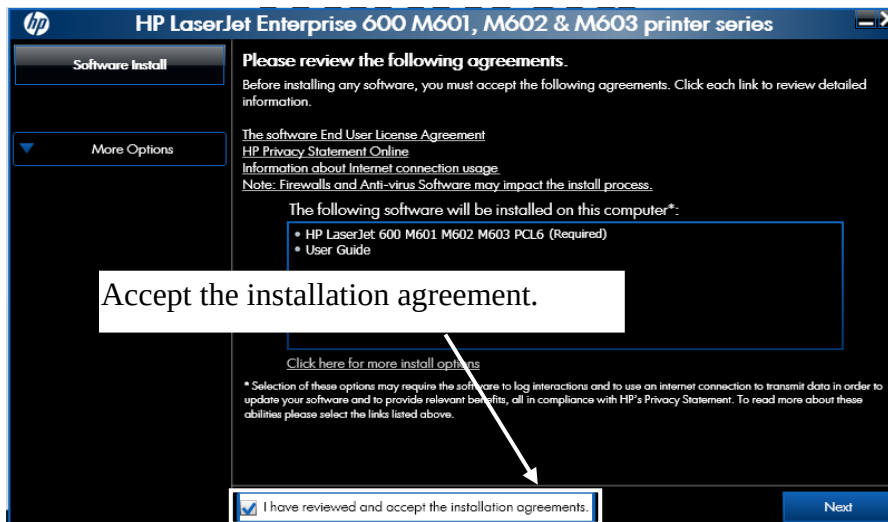
- sadmin help: used to view the commonly used commands;
- sadmin status: used to view the status of McAfee.

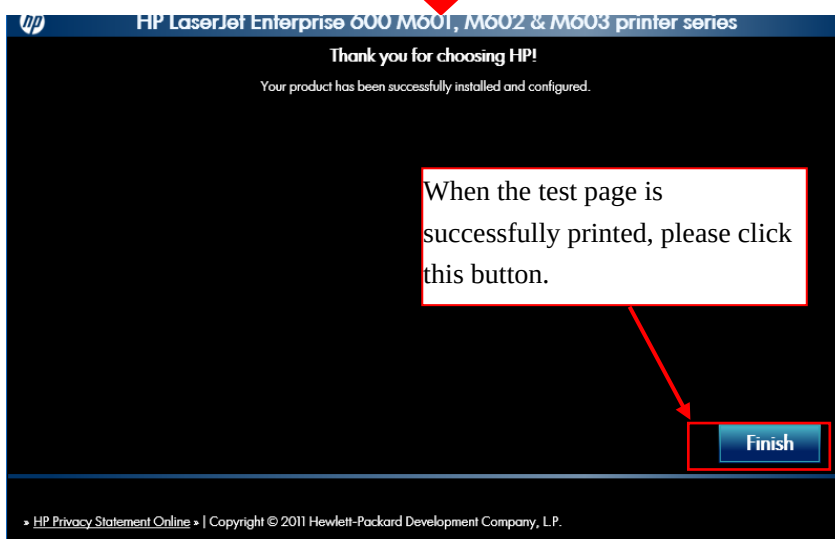
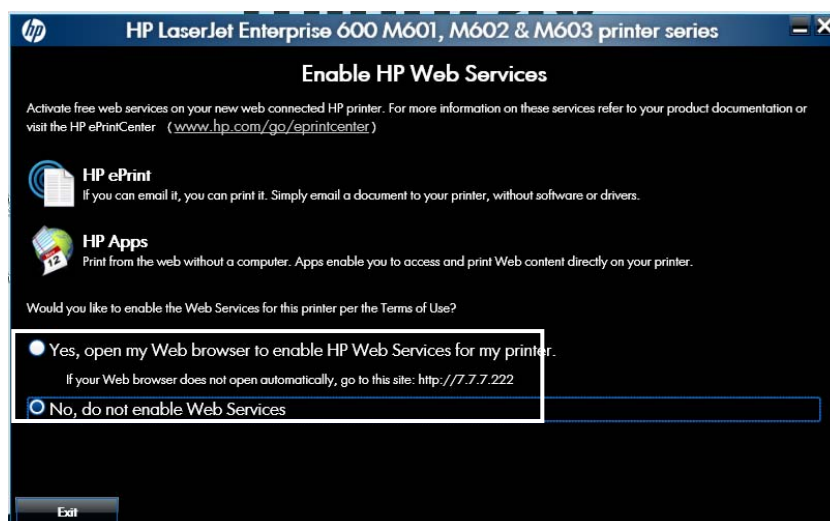
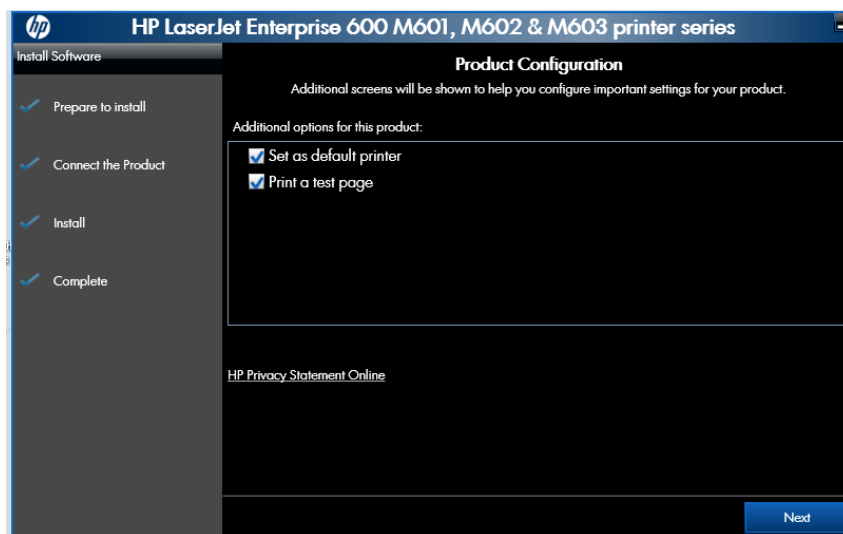
Installing the USB Printer

NOTE

-
- **Make sure that the USB cable of the printer is not connected with computer.**
-

1. Right click the mouse to open CD driver, and then double click "setup.exe" file in the root directory.
2. Install the driver by following the indications shown below in the order of from left to right and from top to bottom:





-
3. After test page is printed, close all the pop up windows and access “Printers and Faxes” again.
 4. Select “HP LaserJet 600 M601 M602 M603 PCL6” printer and re-name it as “M602_usb” to complete installation of printer’s USB driver.

NOTE

- **For HP LaserJet 2035n Printer, Double click the ‘Setup.exe’ which is in the driver CD root directory. Follow the installation guide and choose to install network driver or USB driver as needed.**
 - **When installing a printer, select to install network driver or USB driver based on the actual requirement. Connect the printer to the CMS network if network printing is needed.**
-

3.10.2 Tasks after Printer Installation

To improve the print effect, follow the steps as below:

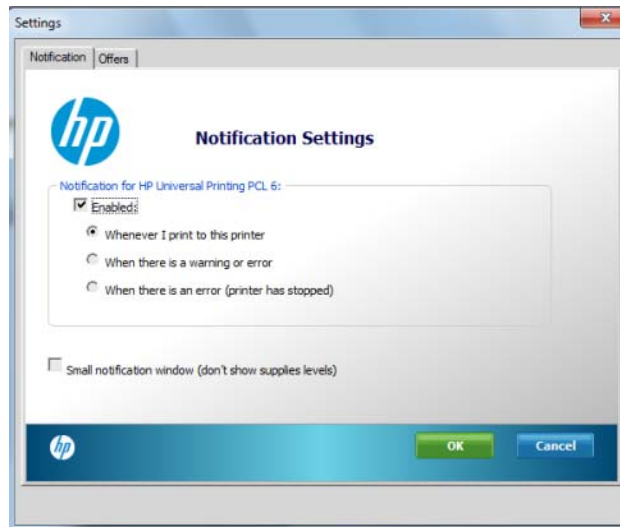
1. Enter the Windows desktop and select “Start” → “Control Panel” → “Hardware and Sound” → “Devices and Printers”.
2. In the “Printers and Faxes” field of the “Devices and Printers” window, select the desired printer and right click the printer.
3. Click the “Printer properties” option from the pop-up menu to display the printer properties window.
4. Click the “Advanced” tab.
5. Unselect the “Enable advanced printing features” option.
6. Click the “OK” button to apply the setting and close the printer properties window.

If printer HP LaserJet M401n and M602 are connected to the CMS, to block the pop-up functional window when running the CMS, follow the steps as below:

1. After install the printer, implement a printing task.
A prompt window displays at the bottom-right of the desktop, as shown in the following figure.



2. Click the “Supplies Details” option to display the “Settings” window, as shown in the following figure.



3. Click the “Notification” tab, and unselect the “Enabled:” check box.
4. Click the “Offers” tab, and unselect the “Allow special device information and offers to be displayed” check box.
5. Click the “OK” button to apply the settings and close the “Settings” window.

3.11 Installing Recorder

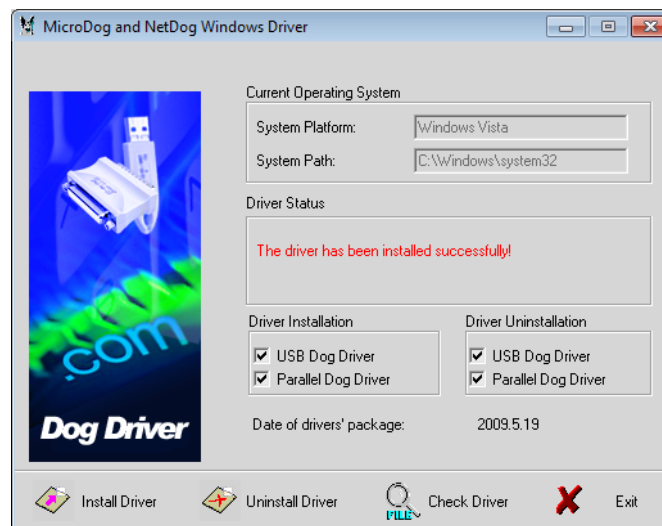
No recorder driver is needed. Recording can function directly after system software is installed.

3.12 Installing USB Dongle Driver

1. Insert the CMS software CD and then execute “MicroDogInstdrv.exe” under the “USB Dog Driver” folder to enter the following window.



2. Set as the figure shows and then select “Install” to enter the following window after completing installation.



3. Select “Exit” when the prompt message “Driver Installation Completed!” appears.

3.13 Installing the CMS Software

3.13.1 Setting the Region and Language of the Operating System

If the language used by the current operating system is just the language the custom wants to display on the CMS, it is unnecessary to set the region and language for the operating system. In this case, just skip this step.

If the language of the operating system is English but the custom wants a non-English for the CMS, it is necessary to set the region and language of the operating system. To set the region and language, follow this procedure:

1. Enter the Windows desktop and select “Start” → “Control Panel” → “Clock, Language, and Region” to enter the “Clock, Language, and Region” window.
2. Click the “Region and Language” option to display the “Region and Language” window.
3. Click the “Format” tab, and then select the desired language (locale) from the “Format” drop-down list.
4. Click the “Location” tab, and then select the desired language (locale) from the “Current location” drop-down list.
5. Click the “Administrative” tab, and then click the “Change system locale...” button to display the “Region and Language Settings” window.
6. Select the desired language (locale) from the “Current system locale” drop-down list.
7. Restart the computer to apply the setting.

3.13.2 Setting Operating System Time

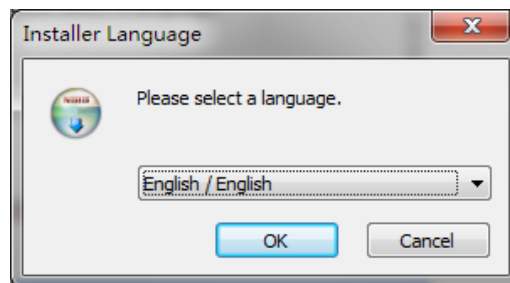
1. Enter the Windows desktop and select “Start” → “Control Panel” → “Date and Time” to display the “Date and Time” window.
The current tab displays the “Date and Time” tab.
2. Click the “Change time zone...” button to display the “Time Zone Settings” window.
3. Select the “Automatically adjust clock for Daylight Saving Time” option.
4. Click the “Internet Time” tab.
5. Click the “Change settings...” button to display the “Internet Time Settings” window.
6. Unselect the “Synchronize with an Internet time server” option.
7. Restart the computer to apply the setting.

3.13.3 Installing the CMS System Software

NOTE

- Insert the USB dongle before installing the system software.
 - The software version of the USB dongle must match the version of CMS.
 - Connect the network cable. The independent network adapter needs to connect to network.
-

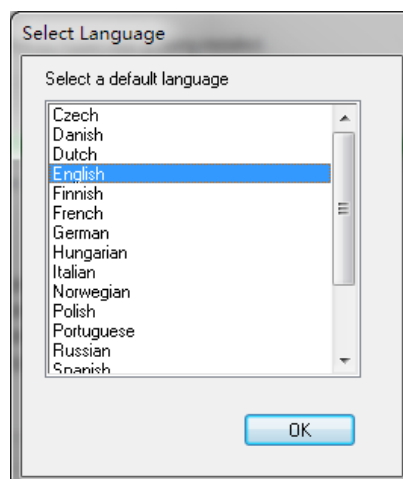
1. In the CMS software CD, double click “Setup.exe” under the “Setup” directory to enter the following window.



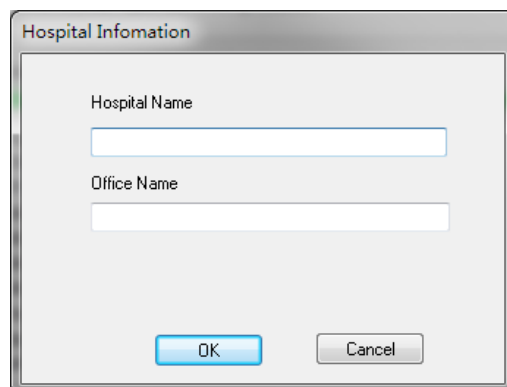
NOTE

- The language selected as shown in the figure above is the one for display on the screen during the process of installation but not the default one when the CMS is operating. The language used when the CMS is operating is to be set up in the following steps.
 - Set OS language to English before installing CMS system software in English or other language operating system. Change to the desired OS language after the installation.
-

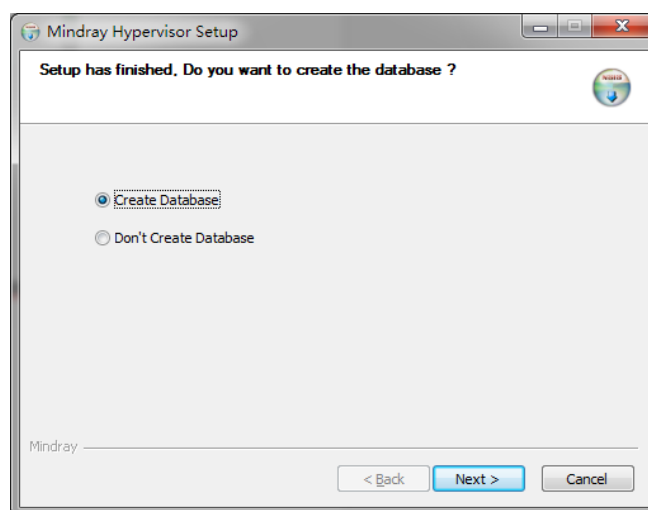
2. After selecting “Ok”, the following window will display.



-
3. Select the desired language from the pop-up box and then click “OK” to enter the following window.

A dialog box titled "Hospital Information" with a light gray background. It contains two text input fields: "Hospital Name" and "Office Name". Below the fields are two buttons: "OK" and "Cancel".

4. Fill in hospital name and department name as required. Then click “OK” to enter the following window.

A dialog box titled "Mindray Hypervisor Setup" with a light gray background. It contains a message: "Setup has finished. Do you want to create the database ?". Below the message are two radio buttons: "Create Database" (selected) and "Don't Create Database". At the bottom, there are three buttons: "< Back", "Next >", and "Cancel".

5. Select the desired installation type as per the following NOTE:

NOTE

- Selecting “Create Database” will destroy the data in the database. “Create Database” is not required in the case of recovering to install CMS software to save the previous data. “Create Database” is required when the CMS software is installed for the first time.
- Select "Don't Create Database" if CMS system software rather than database software is reinstalled and the current database file is compatible with the reinstalled system software.

-
6. When the installation is complete, restart the computer.

After restarting the computer, the CMS will run automatically.

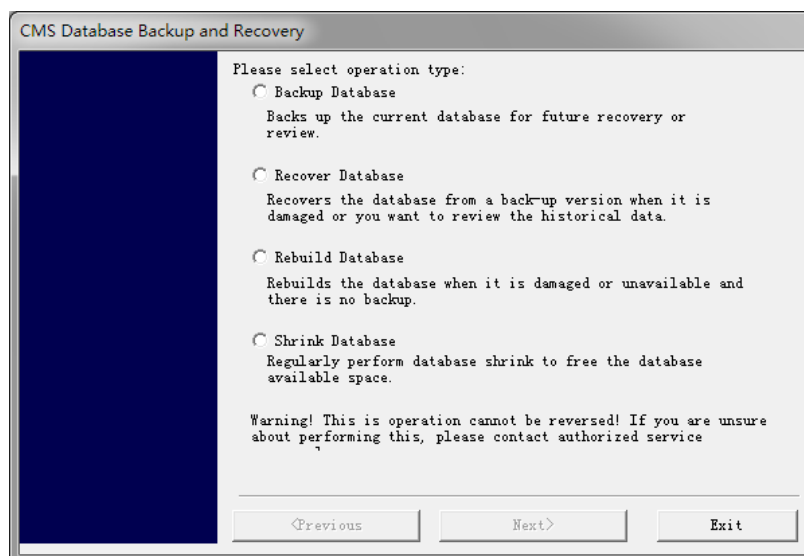
3.13.4 Setting the Size of CMS Screen

Access the CMS system. Select “System Setup” → “Admin Setup” → “Display Setup” → “Screen”. Adjust the screen size of the CMS system in accordance with actual screen size. Restart PC to enable the screen size settings.

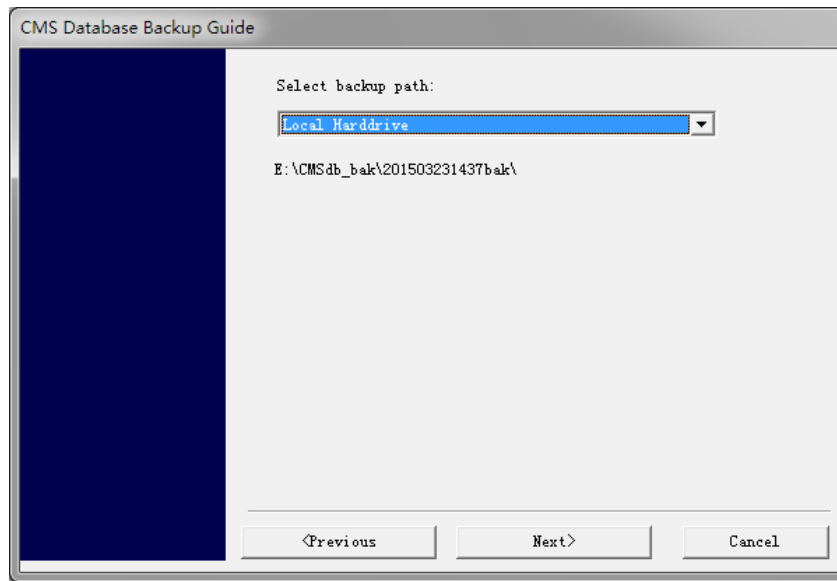
3.13.5 Initial Database Backup

After the CMS is installed, the database needs to be backed up.

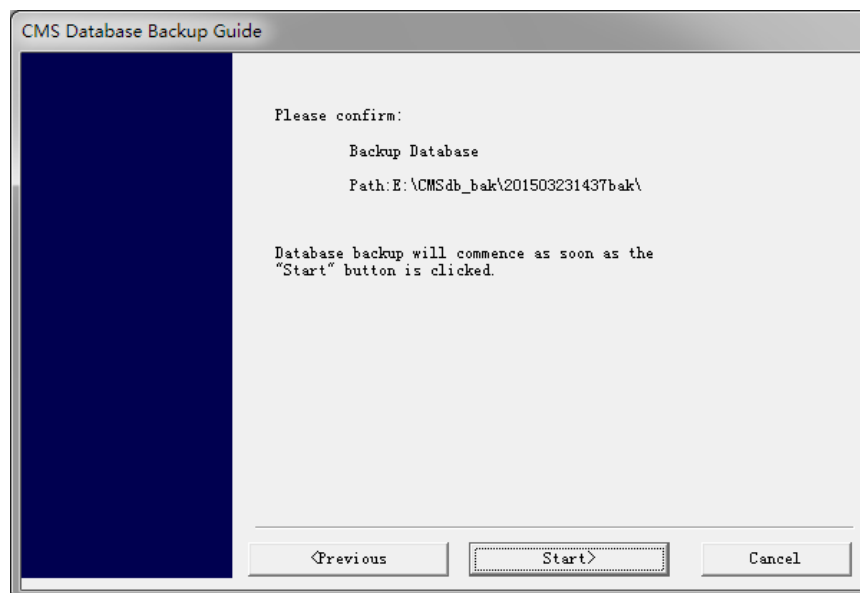
1. Run the CMS. Select “System Setup” → “Admin Setup”(password: 111111) > “Others”, then click “Database Backup and Recovery” to enter the following window.



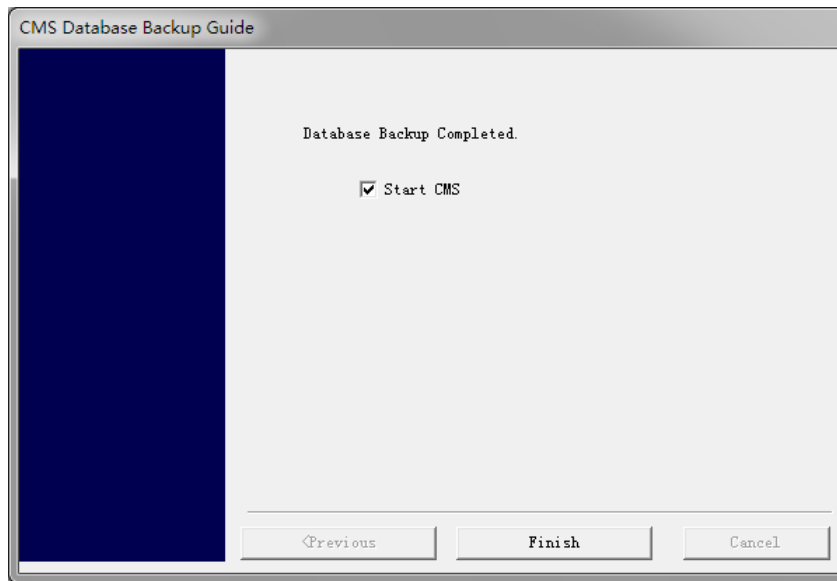
2. Select “Database Backup” to enter the following window:



3. Select “Local Hard disk” and then “Next” to enter the following window.



4. Click “Start” to start to backup the database until the following window appears:



5. Click “Finish” to start the CMS.

NOTE

- The above is initial database backup when the CMS is installed. The database at that time does not contain any data. During actual maintenance, it may be necessary to backup database which has saved a large amount of data. If they are backed up onto the hard disk, the old backup will be deleted and only the latest are backed up.
- When it comes to back up database onto the removable storage medium, make sure that the removable storage medium is not infected with virus. Perform virus scanning or formatting in advance.

3.14 CMS Software Upgrades

If your CMS software is installed in the Windows XP operating system, but now you need to upgrade the CMS software in the Window 7 operating system, you need do the followings to ensure the CMS' running in the Windows 7 operating system:

1. Format C drive, and re-install Windows 7 Professional Embedded Sp1 (32 bit)
2. Change the directory "MySQLData" in D drive to "MySQLData_bak".
3. Run "MySQLInstall.exe" to install MySQL, and then restart the system.
4. Open Windows Task Manager, and end "mysqld.exe" progress.
5. Delete the directory "MySQLData" in D drive, and then change the directory "MySQLData_bak" to be back as "MySQLData". Then restart the system.
6. Install CMS software. After restarting the computer, the CMS can run normally.

NOTE

-
- After installing CMS software, do not perform rebuilding database.
-

3.15 Installing Access Point

3.15.1 Configuring NETGEAR WNDAP 350

Access Point (AP)

3.15.1.1 Login the AP

1. Prepare and power on a PC; and connect the access point with the power supply.
2. Connect the router with the PC through the network cable.
3. Set the PC's IP address and subnet mask. Set IP address to 192.168.0.1 and the corresponding subnet mask to 255.255.255.0. Refer to **3.8 IP Address Setup and Network Connection** for configuring IP address.
4. Open IE browser and input the access point IP address http:// 192.168.0.237. After pressing Enter, the WNDAP350 login screen displays.
5. Input the default username and password. Input "admin" at "Username" field, and "password" at "Password" field.
6. Click "LOGIN". Then the access point's main configuration screen displays.

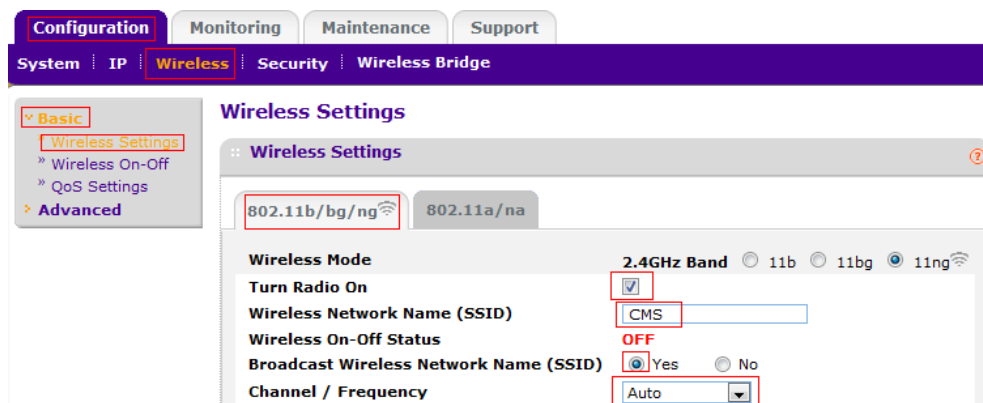
3.15.1.2 Configuring the AP (Without Configuration File)

Step 1: Configuring the Country

1. In the main configuration screen, select “Configuration” → “System” → “Basic” → “General”.
2. Select the proper country or region in the drop-down list. The default country setting is “China”.
3. Click “APPLY” button.

Step2: Enabling AP, Setting SSID and Selecting Channel

1. Select “Configuration” → “Wireless” → “Basic” → Wireless Settings → 802.11b/bg/ng.
2. Set “Wireless Network Name (SSID)” to “CMS” or other proper name.
3. Set “Channel/Frequency” to “Auto”.
4. Set “Broadcast Wireless Network Name (SSID)” to “Yes”.
5. Click “APPLY” button.



NOTE

- The NETGEAR WNDPA 350 access point has enabled a 2.4G channel (Default SSID: NETGEAR_11ng) and a 5G channel (Default SSID: NETGEAR_11a) by default.
- The AP is enabled by default with “Turn Radio On” being selected.

Step 3: Disabling the 5G Channel of AP

Since the current wireless network device does not support the 5G channel, but the 5G channel is enabled by default, this channel should be disabled manually.

1. Select “Configuration” → “Wireless” → “Basic” → “Wireless Settings” → “802.11a/na”.
2. Deselect “Turn Radio On”.

3. Click “APPLY”.
4. Select “Configuration” → “Wireless” → “Basic” → “Wireless On-Off”.
5. Set “Wireless On-Off” to “On”.
6. Click “APPLY” button.

Step 4: Configuring the Authentication Mode

1. Select “Configuration” → “Security” → “Profile Settings” → “802.11b/bg/ng”.
802.11b/bg/ng screen (the 5G channel is not supported temporarily). There are eight SSIDs in 802.11 b/bg/ng (2.4 GHz), with the first SSID being enabled by default.
2. Select “Edit”. The AP’s editing screen displays.
3. Edit the security profiles, show as below:

The screenshot shows the 'Edit Security Profile' configuration page. The 'Profile Definition' section includes fields for 'Profile Name' (NETGEAR), 'Wireless Network Name (SSID)' (CMS), and 'Broadcast Wireless Network Name (SSID)' (Yes). The 'Authentication Settings' section includes 'Network Authentication' (WPA-PSK & WPA2-PSK), 'Data Encryption' (TKIP+AES), and 'WPA Passphrase (Network Key)' (masked). Red circles and arrows highlight the 'Authentication Settings' and 'Network Key Settings' areas.

- ◆ Set “Network Authentication” to “WPA-PSK&WPA2-PSK”.
 - ◆ Set “Data Encryption” to “TKIP+AES”.
 - ◆ Set the Key in “WPA Passphrase (Network Key)”.
4. Click “APPLY” button.

3.15.1.3 Quick AP Configuration (With Configuration File)

Generate a configuration file by using a access point with the configured AP, and use the file to perform the same configuration on the AP of another access point.

Step 1: Preparing the Configuration File

1. Login the access point with the configured AP.
2. In the main configuration screen, select “Maintenance” → “Upgrade” → “Backup Settings”.

3. In the “Backup Settings” screen, select “BACKUP” to save the settings. The default document name: Config.

Step 2: Importing the Configuration File

1. Login the access point to be configured.
2. Select “Maintenance” → ‘Upgrade” → “Restore Settings”.
3. When browsing “Restore saved settings from a file”, select the configuration file of WNDAP350config.
4. Click “APPLY” button.

After importing the configuration file, all the related settings are imported, including IP addresses, subnet mask, SSID, and key. Please modify those setting in the current access point if necessary. Refer to **3.15.1.4 Modifying the IP Address and Subnet Mask** and **3.15.1.4 Modify AP’s SSID and Key**.

3.15.1.4 Modifying the IP Address and Subnet Mask

1. In the main configuration screen, select “Configuration” → “IP” → “IP Settings”.
2. In the “IP Settings” screen, modify the IP address and subnet mask.
3. Click “APPLY” button.

3.15.1.5 Modify AP’s SSID and Key

1. Select “Configuration” → “Security” → “Profile Settings” → “802.11b/bg/ng”. There are eight SSIDs in 802.11 b/bg/ng (2.4 GHz), with the first SSID being enabled by default.
2. Select “Edit” after the SSID. The AP’s editing screen displays.
3. Modify SSID and key. In “Wireless Network Name (SSID)” field, input the desirable SSID; in “WPA Passphrase (Network Key)”field, input the new key.

The screenshot shows the Netgear configuration interface. At the top, there are tabs for Configuration, Monitoring, Maintenance, and Support. Under Configuration, there are sub-tabs for System, IP, Wireless, Security, and Wireless Bridge. The Security tab is selected. On the left, there is a sidebar with 'Profile Settings' and 'Advanced' options. The main area is titled 'Edit Security Profile'. It contains two sections: 'Profile Definition' and 'Authentication Settings'. In the 'Profile Definition' section, the 'Profile Name' is 'NETGEAR', the 'Wireless Network Name (SSID)' is 'CMS', and the 'Broadcast Wireless Network Name (SSID)' is set to 'Yes'. In the 'Authentication Settings' section, the 'Network Authentication' is 'WPA-PSK & WPA2-PSK', the 'Data Encryption' is 'TKIP+AES', and the 'WPA Passphrase (Network Key)' is masked with asterisks. The 'Show Passphrase in Clear Text' option is set to 'No'. Red circles and arrows highlight the 'SSID Settings' and 'Network Key settings' fields.

-
4. Click “APPLY” button.

3.15.2 Configuring Cisco 2602 Access Point (AP)

3.15.2.1 Logging into the AP

Hardware required:

- Cisco 2602 to be configured and corresponding power cable;
- A COM port cable of Cisco 2602;
- A PC with COM ports
- A network cable
- A USB to COM port converter cable (optional, which is required when the used PC has only USB interfaces, but no COM ports).



Connection:

1. Connect the Cisco 2602 power supply,
2. Connect the Ethernet interface of Cisco 2602 to a network interface of the PC with the network cable.
3. Connect the Console interface of Cisco 2602 to a COM port of the PC with the COM port cable of Cisco 2602 (or to a USB interface of the PC with the COM port cable of Cisco 2602 and the USB to COM port converter cable).

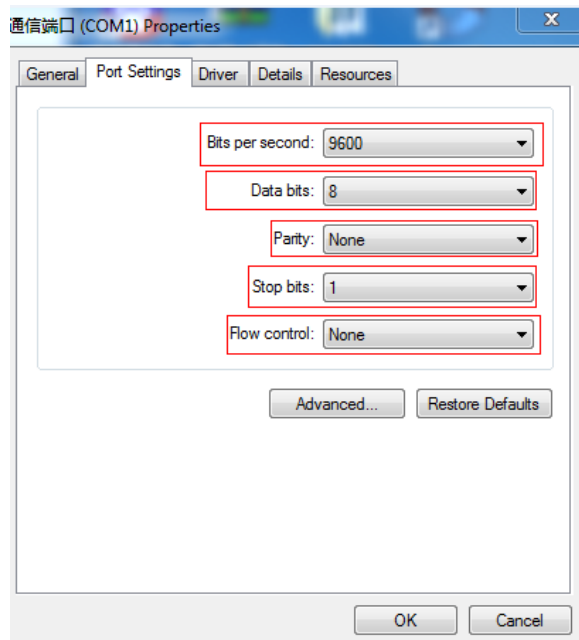
Step 1: Configuring the COM Port Communication Properties

To start HyperTerminal on the Windows operating system: go to “Start” → “All

Programs” → “Accessories” → “Communications” → “HyperTerminal”.

To set COM port communication parameters:

1. In the “Connection Description” screen, enter a name (MR, for example) in the Name field and click “OK”.
2. In the “Connect To” dialog box, select the COM port number (COM1, for example) through which the PC connects to Cisco 2602 access point, and click “OK”.
3. In the pop-up properties dialog box, set the parameters as the Figure shown.



Step 2: Configuring the IP Address and Subnet Mask

1. After COM communication parameters are configured, enter the HyperTerminal main screen.
2. In the HyperTerminal main screen, enter “enable” command to enter the privileged command status.
3. Enter the default password “Cisco”.
4. In the pop-up screen, enter the command “ap# config terminal”.
5. Enter the command “ap{config}# interface BVI1”.
6. Enter the command “ap (config-if)# ip address 196.76.0.2 255.255.255.0” to set the IP address to “196.76.0.2” and the subnet mask to “255.255.255.0”.

Step 3: Accessing the Configuration Screen

1. Set the IP address and subnet mask of the PC. Set the IP address to 196.76.0.1 and set the subnet mask to 255.255.255.0.

-
2. Open the Internet Explorer, and enter the IP address <http://196.76.0.2>. After pressing Enter, A security prompt is displayed.
 3. Enter the default user name and password. Both of them are “Cisco”.
 4. Select the “Remember my credentials” check box.
 5. Click “Ok”. The Cisco 2602 configuration screen displays.

3.15.2.2 Configuring AP (Without a Configuration File)

Step 1: Enabling the AP and Setting Channels

1. On the top of the configuration screen, select “NETWORK”. Then the NETWORK screen displays.
2. On the right of NETWORK screen, select “NETWORK INTERFACE” → “RADIO-802.11n 2G.Hz”.
3. Select “SETTINGS” tab.
4. In the SETTING screen, set the following items:
 - ◆ Select “Enable” for “Enable Radio”.
 - ◆ Select “Access Point” for “Role in Radio Network”.
 - ◆ Select “Least Congested Frequency” for “DefaultRadio Channel”.
 - ◆ Select all channels in the “Least Congested Channel Search” box.
5. Click “Apply” button.

Step 2: Configuring Encryption

1. On the top of the configuration screen, select “SECURITY”.
2. On the right of SECURITY screen, select “Encryption Manager”.
3. Select “AES CCMP + TKIP” for “Cipher”.
4. Click “Apply-Radio” button.

Step 3: Configuring an SSID

1. On the right of SECURITY screen, select “SSID Manager”.
2. Enter “CMS” or other proper name in “SSID”.
3. Select “Radio0-802.11N2.4GHz” for “Interface”.
4. Click the nearest “Apply” button below. The configured CMS appears in the “Current SSID List” box.

Security: Global SSID Manager

SSID Properties

Current SSID List

< NEW >

CMS

SSID: CMS

VLAN: < NONE > Define VLANs

Backup 1:

Backup 2:

Backup 3:

Band-Select: ☐ Band Select

Interface: ☒ Radio0-802.11N 2.4GHz ☐ Radio1-802.11N 5GHz

Step 4: Configuring the Authentication and Key of the SSID

1. In “Current SSID List”, select “CMS” or the name you configured in last step.
2. In “Client Authenticated Key Management” area, select “Mandatory” for “Key Management”.
3. Select “Enable WPA”, and then select “WPA” in the drop-down list.
4. Select “ASCLL” the password format for “WPA Pre-shared Key” and enter a password containing 10 valid digits.
5. Click the nearest “Apply” button, shown as below.

Multiple BSSID Beacon Settings

Multiple BSSID Beacon

☐ Set SSID as Guest Mode

☐ Set DataBeacon Rate (DTIM): DISABLED (1-100)

Apply Cancel

Step 5: Configuring the SSID Broadcast Mode

1. In the “Guest Mode/Infrastructure SSID Settings” area, select “Single BSSID”.
2. Select “CMS” in “Set Single Guest Mode SSID” drop-down list.
3. Click “Apply” button below.

Radio0-802.11N 2.4GHz:

Set Beacon Mode: ☒ Single BSSID ☐ Multiple BSSID

Set Single Guest Mode SSID: CMS

Set Infrastructure SSID: < NONE > ☐ Force Infrastructure Devices to associate only to this SSID

Radio1-802.11N 5GHz:

Set Beacon Mode: ☒ Single BSSID ☐ Multiple BSSID

Set Single Guest Mode SSID: < NONE >

Set Infrastructure SSID: < NONE > ☐ Force Infrastructure Devices to associate only to this SSID

Apply Cancel

3.15.2.3 Quick AP Configuration (With a Configuration File)

Generate a configuration file by using a access point with the configured AP, and use the file to perform the same configuration on the AP of another access point.

Step 1: Preparing a Configuration File

1. Login the access point with the configured AP.
2. On the top of the configuration screen, select “SOFTWARE”.
3. On the right of SOFTWARE screen, select “System configuration”.
4. Right-click config.txt in “Current Startup Configuration File” and save the configuration file to a user-defined address.

The configuration file of the AP is ready for use.

Step 2: Configuring the AP Quickly

1. Login the access point to be configured.
2. Import the configuration.
 - a. On the top of the configuration screen, select “SOFTWARE”.
 - b. On the right of SOFTWARE screen, select “System configuration”.
 - c. Select “Browse” in “Load New Startup Configuration File”.
 - d. Find the configuration file config.txt, and click “Load”.

When the configurations of the original access point are imported, the IP address, subnet mask, SSID and key are imported altogether. If necessary, modify these settings for current access point. Refer to **3.15.2.4 Modify the IP Address and Subnet Mask** and **3.15.2.5 Modify AP's SSID**.

3.15.2.4 Modify the IP Address and Subnet Mask

1. On the top of configuration screen, select “NETWORK”.
2. On the right of NETWORK screen, select “NETWORK INTERFACE” → “IP Address”.
3. Modify the “IP Address” and “IP Subnet Mask” fields.
3. Click “Apply” button.

3.15.2.5 Modify AP's SSID

1. On the top of the configuration screen, select "SECURITY".
2. On the right of SECURITY screen, select "SSID Manager".
2. Select the SSID of the AP to be modified in the "Current SSID List" box.
3. Enter a new SSID in the "SSID" field on the right.
4. Click the nearest "APPLY" button below.

Security: Global SSID Manager

SSID Properties

Current SSID List

< NEW >
CMS

Select the SSID to be modified.

SSID: MR

VLAN: < NONE > Define VLANs

Backup 1:

Backup 2:

Backup 3:

Band-Select: ☐ Band Select

Interface: ☒ Radio0-802.11N 2.4GHz ☐ Radio1-802.11N 5GHz

Network ID: (0-4096)

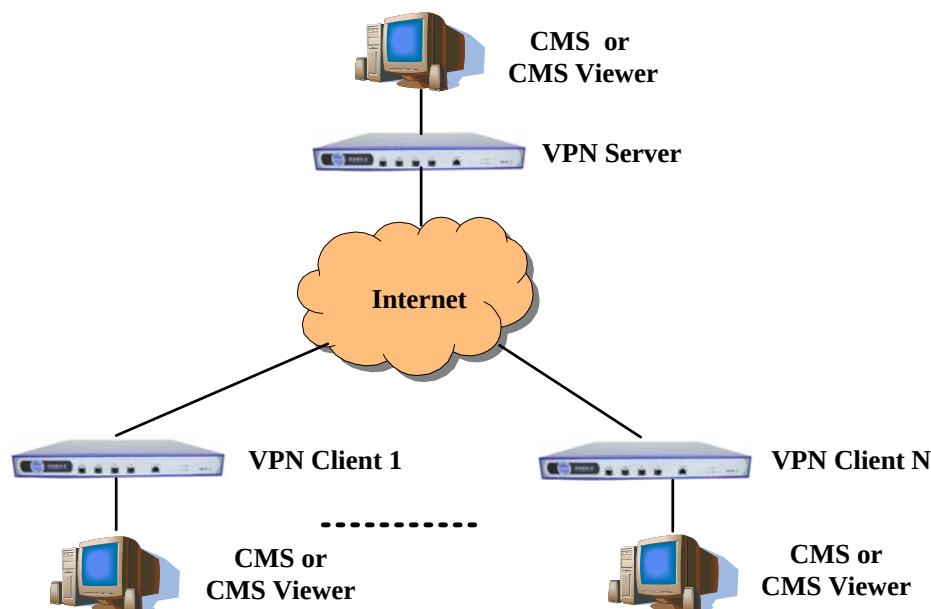
Enter a new SSID.

3.16 Viewing Remote CMSs through VPN

The CMS supports being viewed by other CMSs or CMS viewers through VPN. The VPN network is shown as below: the VPN server requires a fixed IP address through a private line to connect to Internet, while the connection of other VPN clients is as follows:

- If a CMS is viewed by other CMSs, the VPN clients of all the CMSs shall connect to Internet through private lines;
- If a CMS is viewed by a CMS viewer, the VPN client of the CMS shall connect to Internet through a private line, while the VPN client of the CMS Viewer can connect to Internet through either a private line or an ADSL.

Please refer to the accompanying documents for the detailed settings for VPN equipment.



When the VPN network is set up, you can directly connect one of the network interface cards of the CMS with LAN port of VPN, or connect them through switching equipment, and set correct IP address for the CMS.

When you use Remote CMS function to view other CMSs in VPN network, you need to connect other CMSs in Communication Settings menu. Refer to CMS operator's manual for details.

NOTE

- If there are two network cards installed on the host of the CMS, please differentiate the network card for bedside monitor from the card for multiple CMSs / CMS Viewer / other information systems, and set correct IP addresses for the corresponding network cards.
 - If there are two network cards installed on the host of the CMS, please respectively connect them with the corresponding network according to their specific usage.
 - Do not connect two network cards in the same network segment at one time. For example, you can not connect two network cards with bedside monitors' network.
-

FOR YOUR NOTES

4 About Telemetry Monitoring System (TMS)

The configurations about the TMS are operated at the central monitoring system (CMS), such as programming the telemetry packs, admitting the transmitter, discharging the transmitter from the CMS, refer to the *TMS-6016 Installation Guide (P/N: 046-007624-00)*.

FOR YOUR NOTES

5 McAfee Solidcore S3 Control

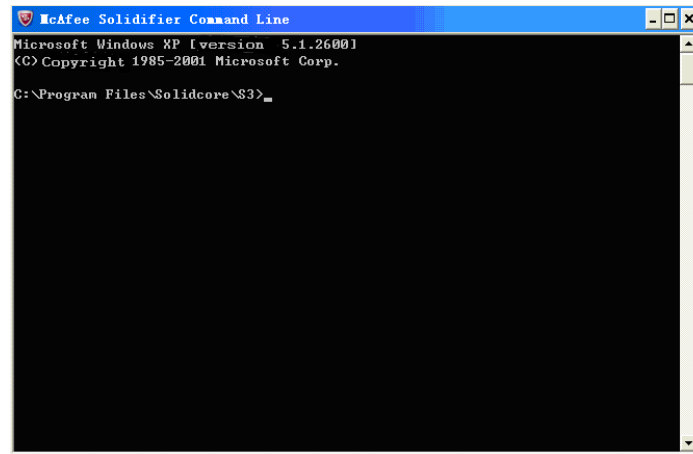
5.1 Installation

For the CMS with Solidcore S3 Control software (Optional), the installation of Solidcore S3 Control will be required.

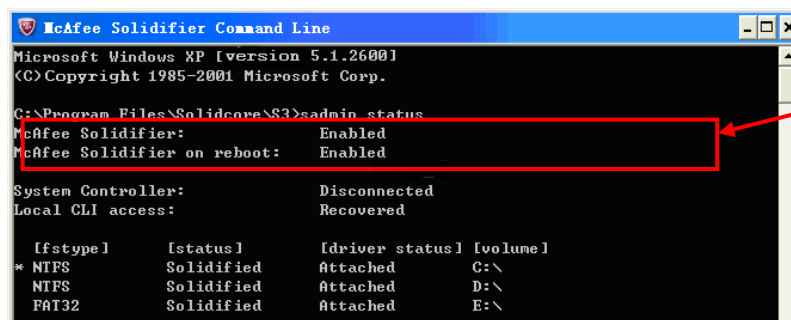
1. Re-start the CMS host, and access the CMS screen.
2. Start “Windows Task Manager”, and then click the “Processes” tab.
3. End the “Master.exe” process, and then close the “Windows Task Manager” window to return the OS desktop.
4. Insert the McAfee Application Control Software CD (P/N: 115-012226-00) into the drive, and then select “SOLIDCOR510-6824_WIN” → “Setup-win2008-win7”.
5. In the “Setup-win2008-win7” folder, right click the “setup-win-7-x86-6.0.0.340.exe” file to display a pop-up menu.
6. Select the “Run as administrator” option to enter the Installation Wizard window.
7. Click the “Next” button.
The “License Agreement” window displays.
8. Select the “I accept the terms in the license agreement” option, and then click the “Next” button.
9. In the “Customer Information” window, input the information as follows:
 - ◆ User Name: input “cmsuser”.
 - ◆ Organization: input “hospital”
 - ◆ Serial Number: input the serial number listed on the Solidcore authorization label.
(The Solidcore authorization label is located on the CD case of the CMS system software)
10. After input the information, click the “Next” button.
11. In the subsequent steps of the installation, the configurations are set by default.

5.2 Enabling the Solidifier

1. Once the installation is complete, double click the icon of *McAfee Solidifier Command Line* on the desktop. Then the following McAfee Command Line window displays.



2. Input “sadmin so” in the command line window, and press the “Enter” key. The computer performs solidification scan.
3. After the scan is done, input “sadmin enable” in the command line window, and press the “Enter” key.
At this moment, a message indicates that Solidcore will run after the computer restarts.
4. Re-start the CMS host after the above operations are finished.
5. Enter the CMS screen, and then press the group keys “Ctrl+Alt+Del” to start “Windows Task Manager”.
6. Click the “Processes” tab, and then end the “Master.exe” process.
7. Close “Windows Task Manager”, and return to the OS desktop.
8. Double click the “McAfee Solidifier Command Line” icon on the desktop to access to the “McAfee Command Line” window.
9. Input the “sadmin status” to check whether the Solidcore is in the “Enabled” state or not, as shown in the following figure.



If the state is correct, it indicates that the installation is complete.

WARNING

- **Ensure that the operating environment of CMS is free from virus before the installation of Solidcore. Otherwise, the CMS may break down after the installation of Solidcore.**
-

NOTE

- **The above-mentioned installation is required only when you select to install the software of Solidcore S3 Control.**
 - **Perform the installation of Solidcore S3 Control after other software is installed.**
-

5.3 Adding or Editing Additional Applications

If applications have to be added, deleted or updated after the installation of Solidcore S3 Control, perform the following to enable the changes under the control of Solidcore:

1. Double click the icon of McAfee Solidifier Command Line on the desktop and the corresponding window will display.
2. Input "sadmin disable" in the command line, press the "Enter" key, and then restart the PC.
The Solidcore control is disabled. You can add or delete the additional applications.
3. Double click the icon of McAfee Solidifier Command Line on the desktop, and then input "sadmin so" to scan all of applications.
4. Input "sadmin enable" in the command line, press the "Enter" key, and then restart the PC.

Other commonly used commands of McAfee Solidifier include:

- sadmin help: used to view the commonly used commands;
- sadmin status: used to view the status of McAfee.

FOR YOUR NOTES

6 System Recovery

6.1 Recovery and Backup in Single Hard Disk Computer

6.1.1 Overview

The recovery tool can recover the system installed on the computer to either of the following states:

- Previously backed up CMS and database
- Windows operating system

6.1.1.1 Recovery Classification

Quick recovery can be from portable hard disk tool or hard disk.

- Recover from hard disk

This recovery method is the recovery from backup files which are saved in the hard disk. This method can be used when both the hard disk and backup files are in good condition.

- Recover from portable hard disk tool:

This recovery method is the recovery from backup files which are saved in the portable hard disk tool. This method can be used when a new hard disk is used and partitioned already. The requirement for partition is that disk C and D are in the NTFS format and disk E is in the FAT32 format.

6.1.1.2 Application Scope

Before using, it is necessary to determine the cause of system failure. If the computer's failure is caused by the damaged hard disks, you must replace the hard disk and then adopt the method of recovering from portable hard disk tool.

If the system failure is caused by software (including the operating system and CMS software) but the partition of the hard disk still meets the factory requirements, you can recover the system with the method of recovery from hard disk.

NOTE

- **Do not recover the system with the method of recovery from hard disk if system failure is caused by other reasons.**
-

6.1.1.3 Precautions

All files used for hard disk recovery are saved on the Disk E. Users are not allowed to do any operation on the Disk E, such as:

- Deleting a file
- Adding a file
- Modifying a file name
- Formatting
- Changing the partition format. The Disk E must be in FAT32 format, and Disk C and Disk D must be in NTFS format.
- Changing the hard disk partition information, including drive letter for each partition, format, or size of partition.

If system recovery failure is caused by disk information changes, you must re-install the operating system and the CMS.

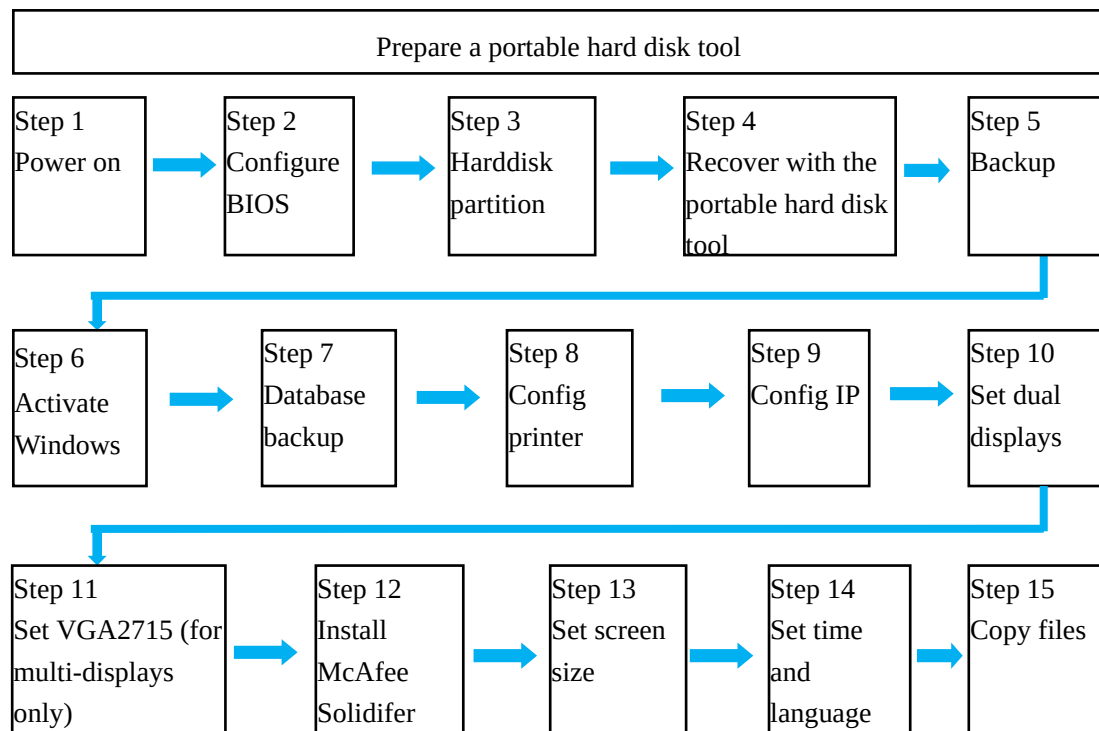
If system recovery failure is caused by damaged files under the ghofiles folder in disk E, you can first delete the ghofiles folder (including the files under it) and then recover the system using recovery disk. If recovery fails, consider to recover only the Windows operating system. If recover still fails, you must re-install the operating system and CMS software.

NOTE

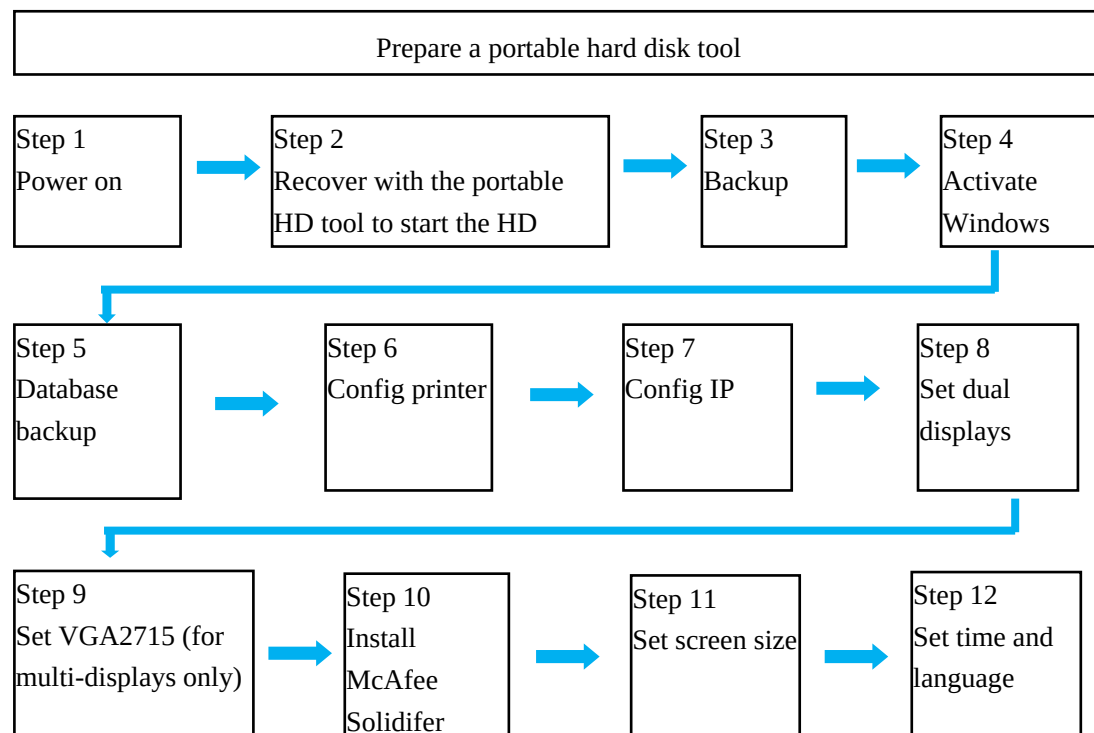
- **System recovery will format disk C and disk D of the hard disk. So backup your desired data in disk C and disk D before performing the recovery operation. We assume no responsibility for loss of data during the recovery.**
-

6.1.2 Recovery Procedure

Recovery from Portable Hard Disk Tool



Recovery from Hard Disk



6.1.3 Creating a Portable Hard Disk

6.1.3.1 Preparation

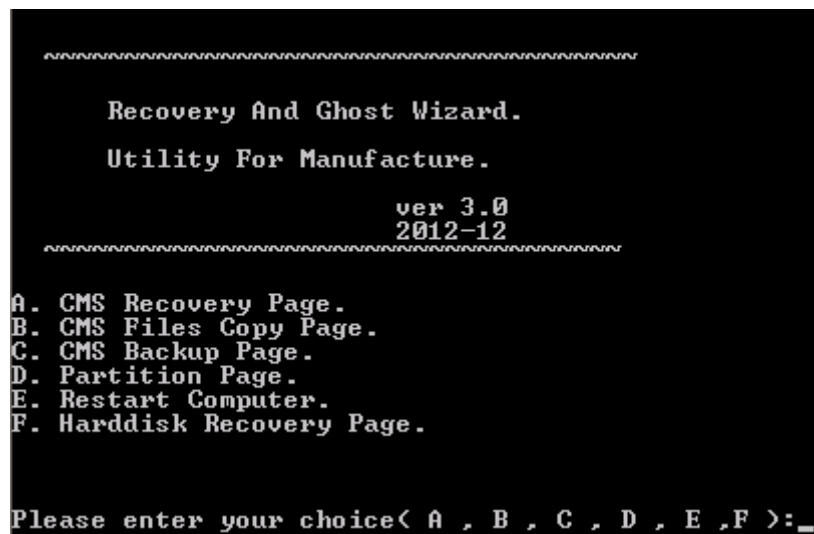
Before installing the software, ensure that the following tools and materials are available:

1. A PC which is pre-installed with a CD-ROM driver and supports USB devices.
2. A portable hard disk (the capacity is equal to or higher than 40 GB) and a USB data cable
3. The CMS tool disk (P/N: 300B-30-47653)
4. The CMS GHOST files of the G-300B-S01 CMS tool (USB version) have been copied to the PC hard disk.

6.1.3.2 Software Installation Process

Formatting the Portable Hard Disk

1. Properly connect the hard disk to the PC.
2. Power on the PC and insert the CMS tool disk (P/N: 300B-30-47653) into the CD-ROM drive of the PC.
3. Set the PC to boot from the CD-ROM drive. After the system starts from the CD-ROM drive, the following window is automatically displayed:



4. Press **Ctrl+C**, and then enter "Y". The system enters the DOS command line window.
5. Check the portable disk No., and in the command line window, enter "gdisk" and press **Enter**. The partition information of all disks on the PC is displayed, as shown in the following figure:

Disk	Partitions	Cylinders	Heads	Sectors	Mbytes	Model
1	7	19457	255	63	152627.8	MAXTOR STM3160215AS
2	2	19457	255	63	152627.8	HITACHI

Hard disk ID

Hard disk capacity

Hard disk model

As shown in the above figure, currently, two hard disks are mounted on the PC. One hard disk has seven partitions, and the other hard disk has two partitions. The portable hard disk needs to be partitioned and formatted before software is installed on the portable hard disk. Therefore, the portable hard disk must be identified first. You can identify the portable hard disk according to the capacity and model of the hard disks. For example, the model of the portable hard disk is HITACHI, so that the system identifies that the disk ID 2 is the portable hard disk and partitions and formats the second hard disk.

NOTE

- The data on the identified hard disk will be deleted in the process, therefore, to avoid unwanted data loss on the PC disk, do ensure that the desired portable disk is identified.

6. Partition and format the portable hard disk.

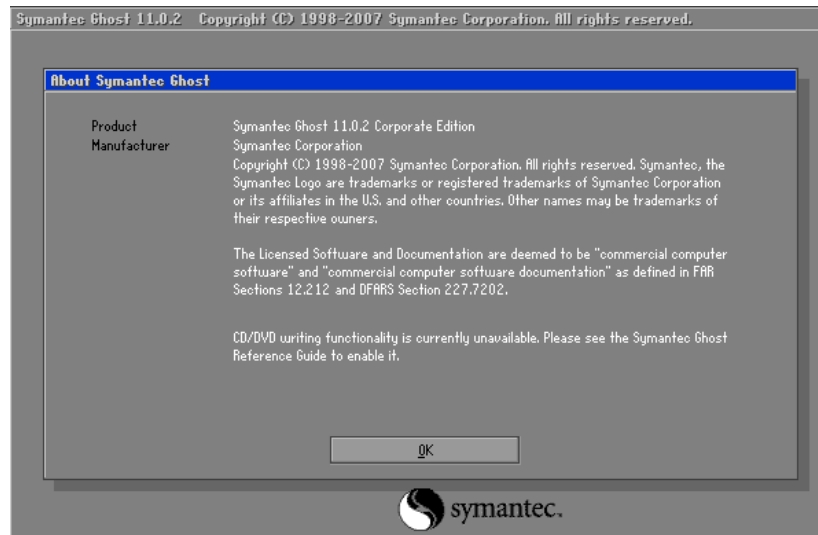
NOTE

- In the following command lines, “□” indicates a space, and thus you need input a space to replace “□” when entering a command; “2” indicates the disk No. of the portable hard disk, and the disk No. must comply with the actual one.
- ◆ Enter “GDISK□2□/DEL□/ALL” and press Enter key. All partitions on the second disk are deleted.
 - ◆ Enter “GDISK□2□/CRE□/PRI□/SZ:600□/FOR□/Q□/V:CMS” and press Enter. Create the first partition on the second disk and set the first partition as the primary partition, with a capacity of 600 MB. Convert the primary partition to FAT32 format and name the volume as “CMS”.
 - ◆ Enter “GDISK□2□/ACT□/P:1” and press Enter key. The first partition of the second disk is activated as the primary partition .
 - ◆ Enter “GDISK□2□/CRE□/EXT” and press Enter key. The system starts to create the extended partition of the second disk.
 - ◆ Enter “GDISK□2□/CRE□/LOG□/FOR□/Q□/V:GHOST” and press Enter key. Create the second partition on the second disk and set the second partition as the logical partition, with a capacity of the remaining space. Format the logical partition in the FAT32 format and name the volume as“GHOST”.

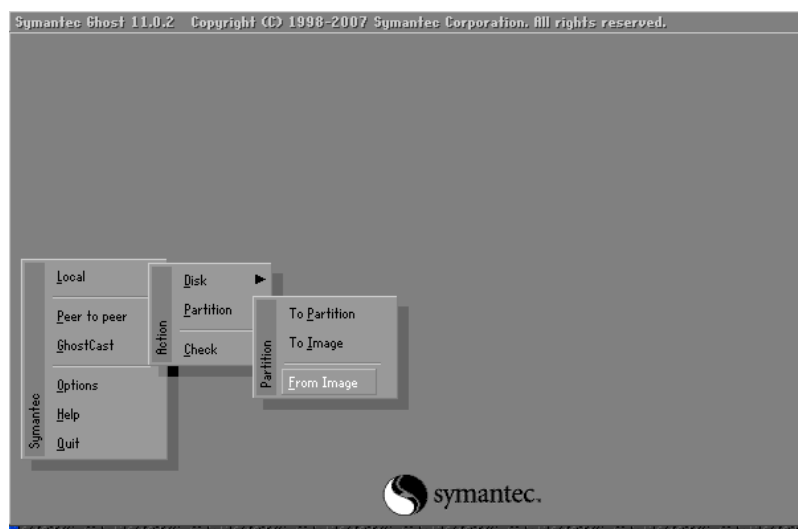
Till now, the portable hard disk is partitioned and formatted.

Installing Software on the Portable Hard Disk

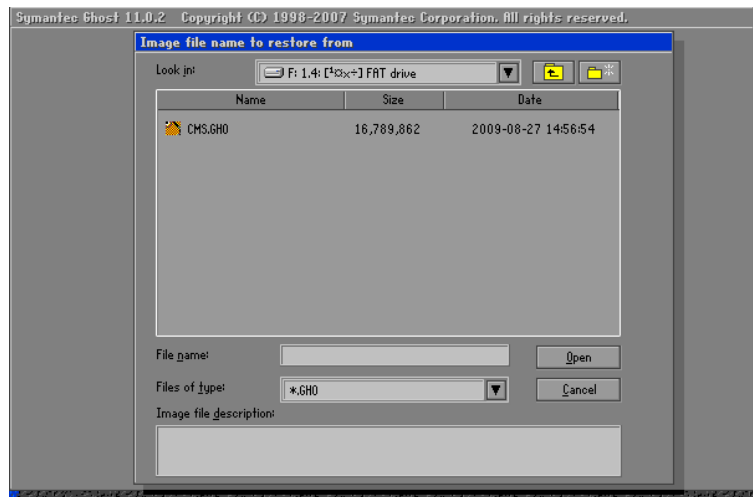
1. In the DOS command line window, enter “GHOST” and press Enter. The GHOST window displays, as shown below.



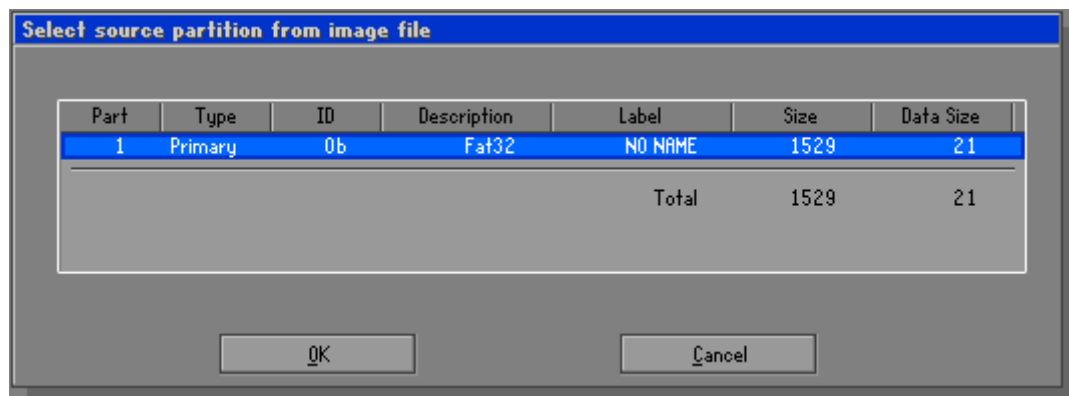
2. Click “OK” or press “Enter”. The GHOST main screen displays, as shown below.



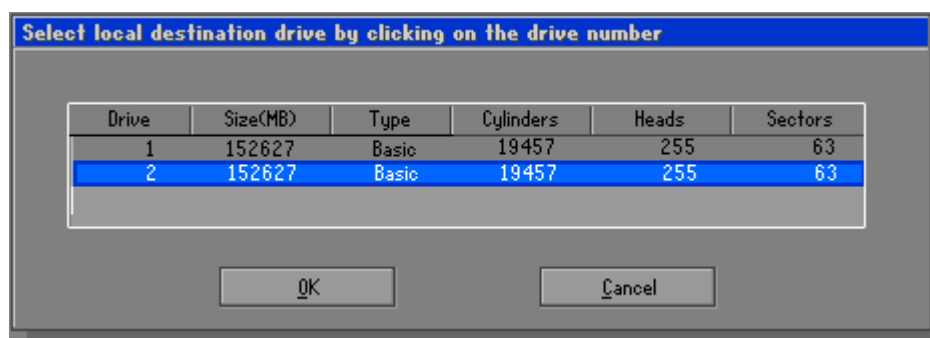
3. Choose “Local” → “Partition” → “From Image” by using the mouse or keyboard. The following screen displays:



4. In the storage path, select the CMS.GHO file as the source file for image recovery. The following screen displays:



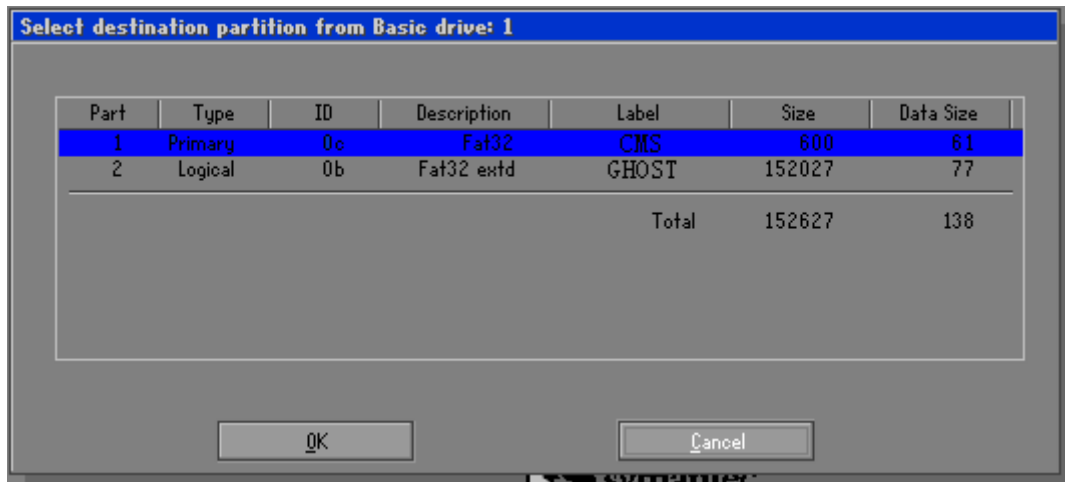
5. Select the source partition of the to-be-recovered image file and press **Enter** or click “OK”. The following screen displays:



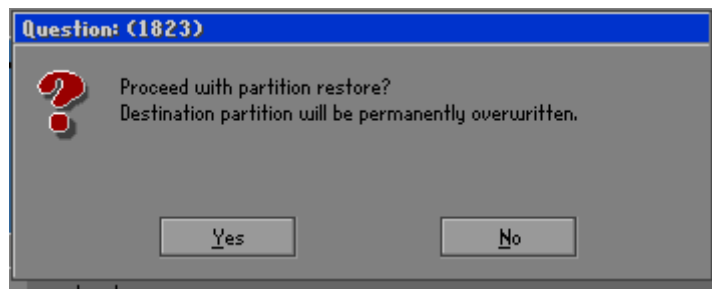
6. Select the target drive for GHOST recovery by using the mouse or keyboard. In this case, select the second drive and click “OK”. The following screen displays.

NOTE

- The portable hard disk has been divided into two partitions when the portable hard disk is formatted. If there are not two partitions in the following screen, the target drive is not the correct one. In this situation, click “Cancel” and select the correct target drive on the main GHOST screen.
-



7. Select the partition where the software is to be installed as the target partition. In this case, select the first partition and click “OK”. In the following dialog box, click “Yes”. Then, the system starts the recovery.



8. After the recovery is completed, select “Reset Computer” in the displayed dialog box.
9. After the PC is restarted, create a folder named “GHOST” in the second partition (the volume name is **GHOST**) on the portable hard disk.

Till now, the CMS tool (USB version) is completely installed.

Copying the Backup Folders to the Portable Hard Disk

The Shenzhen headquarter provides the backup folder (**EN-WIN-R**) for the Windows system of the dual-hard disk host and the backup folder (**EN-CMS-R**) for the Windows system and the BeneVision CMS system of the dual-hard disk host. Copy the **EN-WIN-R** and **EN-CMS-R** folders to the GHOST folder in the second partition (the volume name is **GHOST**) on the portable hard disk. In this way, the portable hard disk for recovering the Windows system or the Windows system and the BeneVision CMS system is prepared.

6.1.4 Configuring BIOS

1. Set BIOS password to “Hypernet” (case sensitive).
2. Set “SATA Emulation” to “RAID”.
3. Set start mode to starting from hard disk first.

6.1.5 Hard Disk Partition

The hard disk must have three (3) partitions:

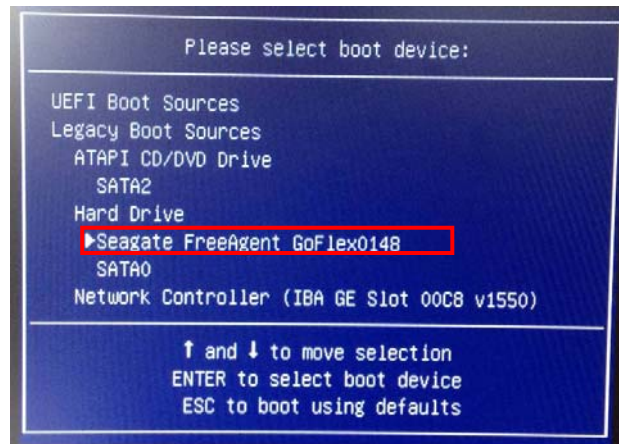
- Disk C: greater than 60 G, NTFS format
- Disk D: greater than 380 G, NTFS format
- Disk E: greater than 60 G, FAT32 format

NOTE

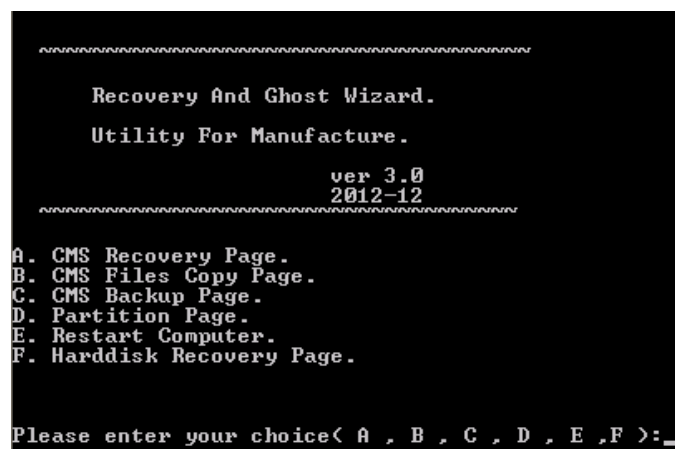
-
- **The file system of the disk partition where the data of the database is located must be in the NTFS format.**
 - **System partitioning will format the whole hard disk and delete all data. Take care when performing system partitioning. Make sure that all useful data are already backed up.**
-

To partition the hard disk, do the followings:

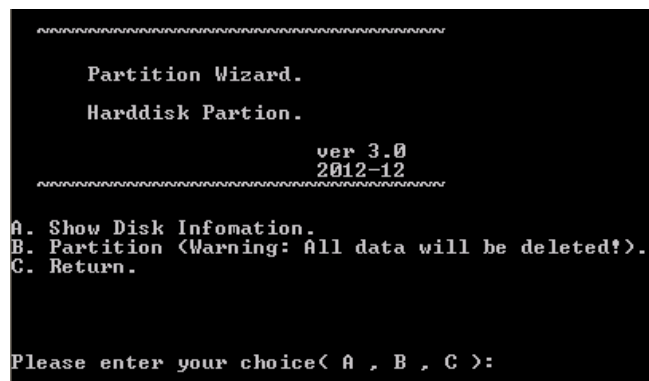
1. Connect the portable hard disk to the computer host.
2. Press F9 key on the keyboard. Then the following screen displays.



3. Select the automatically recognized choice. Then the following screen displays.



4. Select “D” to enter partition screen.



5. Input “B”. Then a message is shown for your confirmation.
6. Input “Y” (case insensitive). The system starts partition.

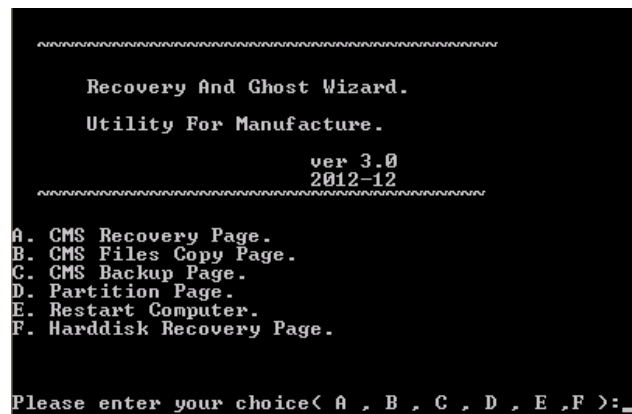
6.1.6 Recovery with Hard Disk

System backup files are saved in Disk E. They are:

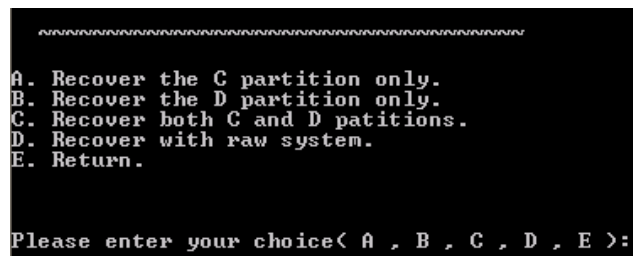
- E:\EN-WIN-R (Windows operating system backup)
- E:\EN-CS-R (the BeneVision CMS backed up last time)

To recover the hard disk:

1. Connect the portable hard disk tool to the computer host.
2. Restart the computer. Wait until the main menu displays.



3. In the main menu, input “F” to enter hard disk recovery wizard, as shown below.



4. Input “A”, “B”, “C” or “D” (case insensitive). Then a message is shown for your confirmation.
 - ◆ Select “A” to recover the Disk C only. The operating system and CMS are recovered.
 - ◆ Select “B” to recover the Disk D only. The database files are recovered.
 - ◆ Select “C” to recover both Disk C and Disk D. The operating system, CMS and database files are recovered.
 - ◆ Select “D” to recover the original operating system. CMS and database files are not installed.

NOTE

- If Disk C and Disk D are recovered simultaneously, the data in the CMS database will be lost and the system will recover the factory configuration. Before recovery, make sure that files in the database are properly handled.
 - If the operating system is recovered, only data in Disk C is recovered.
-

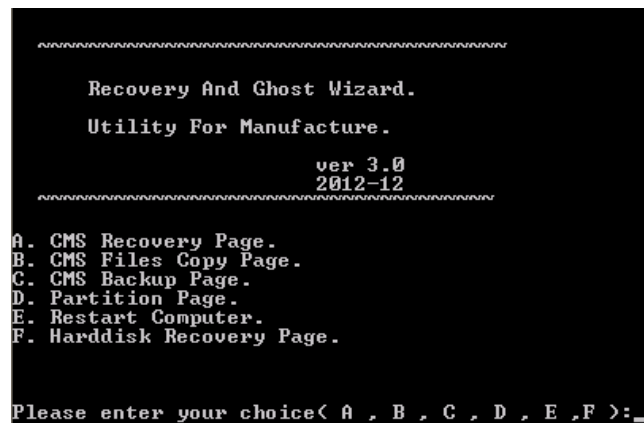
5. Input “Y” (case insensitive). The system starts recovery.
6. Wait until the main menu displays. Then input “E” to restart the computer.
7. Remove the portable hard disk tool from the computer.
8. Restart the computer.

6.1.7 Recovery with Portable Hard Disk

NOTE

- Make sure that the backup files in the CD are correct in terms of computer type, single/double screen and Windows operating system version.
-

1. Connect the portable hard disk to the computer host.
2. Restart the computer. Wait till the following main menu displays:



```

Recovery And Ghost Wizard.
Utility For Manufacture.

ver 3.0
2012-12

A. CMS Recovery Page.
B. CMS Files Copy Page.
C. CMS Backup Page.
D. Partition Page.
E. Restart Computer.
F. Harddisk Recovery Page.

Please enter your choice< A , B , C , D , E ,F >:_
```

3. Input “A” to enter the recovery wizard screen. .

```

A. Recover Chinese Windows OS.
B. Recover Chinese CMS.
C. Recover Chinese CMS with Printer Driver.
D. Recover English Windows OS.
E. Recover English CMS.
F. Recover English CMS with Printer Driver.
G. Recover OEM CMS.
H. Recover OEM CMS with Printer Driver.
I. Recover Chinese Windows OS(RAID Mode).
J. Recover Chinese CMS(RAID Mode).
K. Recover Chinese CMS with Printer Driver(RAID Mode).
L. Recover English Windows OS(RAID Mode).
M. Recover English CMS(RAID Mode).
N. Recover English CMS with Printer Driver(RAID Mode).
O. Recover OEM CMS(RAID Mode).
P. Recover OEM CMS with Printer Driver(RAID Mode).
Q. Return

Please enter your choice: _

```

4. Input “L” or “M” (case insensitive). Then a message is shown for your confirmation.
 - a. Select “L” to recover the operating system.
 - b. Select “M” to recover the operating system and the central station.
5. Input “Y” (case insensitive). The system starts recovery.
6. Wait until the following screen displays. Then press any key to restart the computer.

```

#####

      XXX has been installed successfully!!!
      Press any key to restart computer!!!

#####

Press any key to continue...

```

7. Remove the portable hard disk tool from the computer.
8. Restart computer.

6.1.8 System Backup

NOTE

- **System backup will back up the data in both C and D partitions simultaneously and overwrite the previous system backup.**

1. Connect the portable hard disk tool to the computer disk.
2. Restart the computer. Wait till the following main menu displays:

```

Recovery And Ghost Wizard.
Utility For Manufacture.

ver 3.0
2012-12

A. CMS Recovery Page.
B. CMS Files Copy Page.
C. CMS Backup Page.
D. Partition Page.
E. Restart Computer.
F. Harddisk Recovery Page.

Please enter your choice( A , B , C , D , E ,F ):_

```

3. Input "C" to enter the backup wizard screen.

```

A. Recover Chinese Windows OS.
B. Recover Chinese CMS.
C. Recover Chinese CMS with Printer Driver.
D. Recover English Windows OS.
E. Recover English CMS.
F. Recover English CMS with Printer Driver.
G. Recover OEM CMS.
H. Recover OEM CMS with Printer Driver.
I. Recover Chinese Windows OS(RAID Mode).
J. Recover Chinese CMS(RAID Mode).
K. Recover Chinese CMS with Printer Driver(RAID Mode).
L. Recover English Windows OS(RAID Mode).
M. Recover English CMS(RAID Mode).
N. Recover English CMS with Printer Driver(RAID Mode).
O. Recover OEM CMS(RAID Mode).
P. Recover OEM CMS with Printer Driver(RAID Mode).
Q. Return

```

4. Input "L" or "M" (case insensitive). Then a message is shown for your confirmation.
 - a. Select "L" to backup the operating system. The backup files are stored in the folder "EN-WIN-R" in Disk E.
 - b. Select "M" to backup the operating system and the central station. The backup files are stored in the folder "EN-CMS-R" in Disk E.
5. When the files are backed up, the following screen displays:

```

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXX GHOST file has been made successfully!!!
Press any key to verify the file!!!

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Press any key to continue...

```

6. Press any key. The Ghost screen displays. The system will automatically start to verify the backed-up files. When the verification is complete, the following screen displays:

```

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Backup completed!!!
Press any key to return to main menu!!!

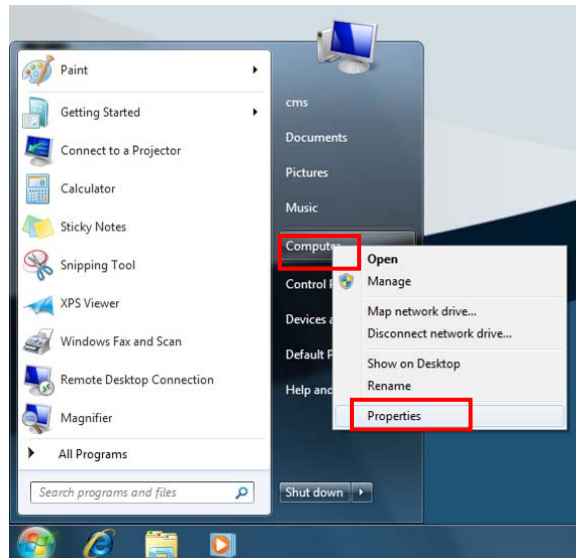
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Press any key to continue...

```

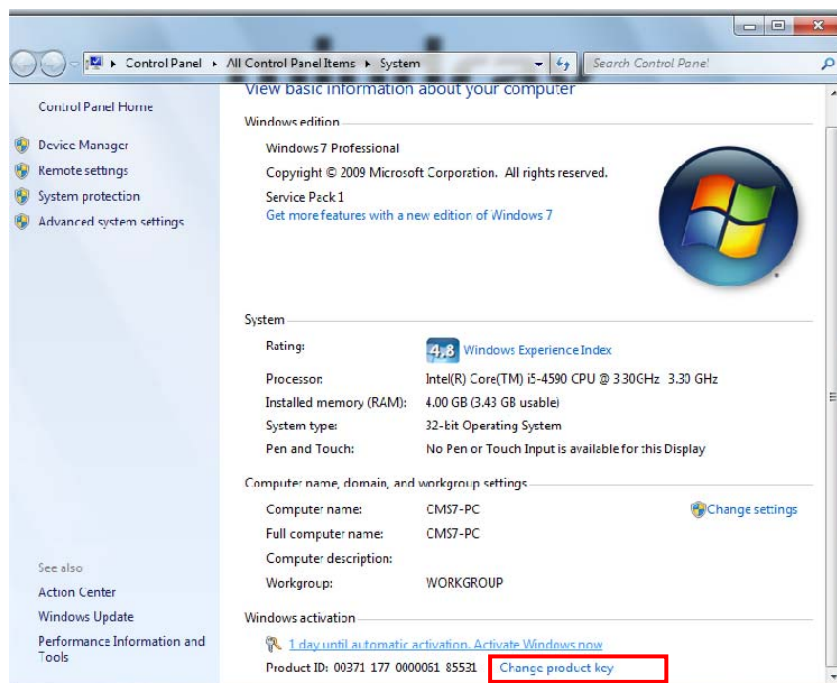

-
7. Press any key to go to the main menu.

6.1.9 Windows System Activation

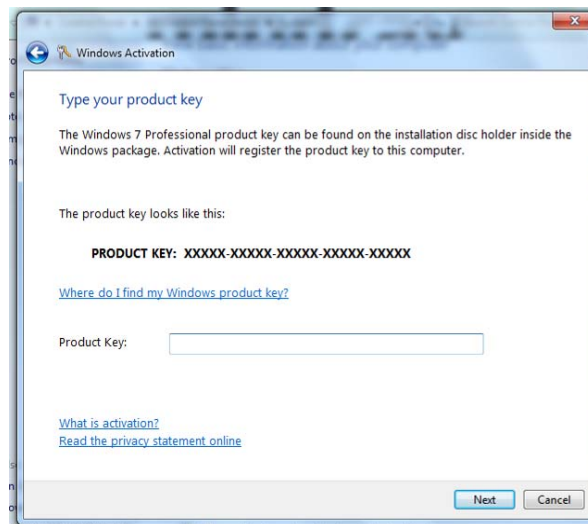
1. Select “Start” button. Right click “Computer”, and then select “Properties”.



2. In the pop-up screen, select “Change product key”.



3. In the pop-up Windows Activation screen, select “Change product key”.
4. In the pop-up Windows Activation screen, input the product key, and then select “Next”.



5. In the pop-up Windows Activation screen, select “Use the automated phone system”, and then complete the activation according to the voice prompt.

NOTE

- If the CMS backup files are recovered, you need to back up the database of the CMS. For details, refer to *3.13.5 Initial Database Backup*.
- After the Windows operating system is recovered, refer to *Chapter 3 System Installation* to install the CMS.

6.1.10 Database Backup

Refer to *section 3.13.5 Initial Database Backup*.

6.1.11 Configuring Printer

Refer to *section 6.1.11 Installing Printer (Optional)*.

6.1.12 Configuring IP Addresses

Refer to *section 3.8 IP Address Setup and Network Connection*.

6.1.13 Installing Dual Displays

Refer to *section 3.7.2 Implement Double Screen Display by DP-to-VGA Adapter* for dual displays connection and settings.

6.1.14 Installing Multiple Displays

Refer to *section 3.7.6 Installing Multi-Screen Extend Device (SUNIX VGA2715)* for three or four displays connection and settings.

6.1.15 Installing McAfee Solidifier

Refer to *Chapter 5 McAfee Solidcore S3 Control*.

6.1.16 Setting the Size of CMS Screen

Access the CMS system. Select “System Setup” → “Admin Setup” → “Screen Size”. Adjust the screen size of the CMS system in accordance with actual screen size. Restart PC to enable the screen size settings.

6.1.17 Setting Language and Time

Refer to *section 3.13.1 Setting the Region and Language of the Operating System* for language settings.

Refer to *section 3.13.2 Setting Operating System Time* for time settings.

6.1.18 Copying files

Copy the files in Ghost folder of Ghost Disk on the portable hard disk tool to the computer E drive root directory.

6.2 Recovery and Backup in Dual Hard Disk Computer

When one of the dual hard disks fails, replace the faulty disk with a new one. Then the new disk will be recovered automatically.

When both hard disks fail, refer to *section 6.1 Recovery and Backup in Single Hard Disk Computer* for recovery.

Refer to *8.10 Dual Hard Disks Failure* for the dual hard disk failure and handling.

FOR YOUR NOTES

7 Maintenance and Cleaning

7.1 Maintenance

Failure on the part of the responsible hospital or institution employing the use of the central monitoring system to implement a satisfactory maintenance schedule may cause undue equipment failure and possible health hazard.

WARNING

- **The safety checks or servicing involving any disassembly or decomposition of devices should be performed by professional servicing personnel; otherwise, it may lead to undue equipment failure and possible health hazards. Turn off the CMS if no patients are to be monitored. If the system has been running for half a year continuously, restart the system.**
-

7.1.1 General Inspection

Whenever your system is repaired, upgraded or has been used for 6-12 months, a thorough inspection should be performed by qualified service personnel to ensure the reliability. Before the central monitoring system is put into use and when it is in use, follow these guidelines to inspect it:

- Inspect the equipment and its accessories for mechanical damage;
- Inspect if the environment and power supply meet the specified requirements;
- Inspect all power cords and signal lines for fraying or other damages, and if they are properly connected and insulated;
- Inspect if the sound system functions normally;
- Inspect if each function of the system is in good condition;
- In case of any damage or abnormality, do not use the central monitoring system. Contact the hospital biomedical engineers or our service personnel immediately.

7.1.2 System Performance Test

After the central monitoring system is reinstalled or quick recovery installation is performed, follow this procedure to make sure that the system functions correctly:

- Display setup
- Functional test of admitting patient
- Record and print Functional test of Dongle
- Clear database
- Shut down

7.2 Cleaning

Your equipment should be cleaned on a regular basis. If there is heavy pollution in your place or your place is very dusty and sandy, the equipment should be cleaned more frequently. The equipment to be cleaned includes the main unit (including power fan, cabinet fan or other fan etc.), displays, printer, recorder, keyboard and mouse. Before cleaning the equipment, consult your hospital's regulations for cleaning, disinfecting and sterilizing equipment

WARNING

- **Be sure to shut down the system and disconnect all power cords from the outlet before cleaning the equipment.**
 - **Cooling fans installed on CPU, display adapter and cabinet should be cleaned on a regular 6-months time interval. If any malfunction of cooling fan is encountered, please contact supplier for replacement as soon as possible. Dust accumulation on cooling fans can lead to fan damage and failure, so that key component may be burned due to high temperature.**
-

The exterior surfaces of the equipment may be cleaned with a clean and soft cloth, sponge or cotton ball, dampened with a non-erosive cleaning solution. Drying off excess cleaning solution before cleaning the equipment is recommended. Following are examples of cleaning solutions:

- Hydrogen peroxide (3%)
- Ethyl alcohol (70%)
- Workstation/server cleaning solutions
- Liquid crystal display (LCD) detergent

Follow these rules to clean the equipment; otherwise, it may melt, distort, or dull the finish of the case, blur lettering on the labels, or cause equipment failures.

- ALWAYS dilute the solutions according to the manufacturer's suggestions.
- ALWAYS wipe off all the cleaning solution with a dry cloth after cleaning.
- NEVER SUBMERGE the equipment into water or any cleaning solution, or POUR or SPRAY water or any cleaning solution on the equipment.
- NEVER permit fluids to run into the casing, switches, connectors, or any ventilation openings in the equipment.
- NEVER use abrasive materials and erosive or acetone –based cleaners.

 WARNING

- **Disinfection or sterilization may cause damage to the equipment; therefore, when preparing to disinfect or sterilize the equipment, consult your hospital's infection controllers or professionals.**
 - **The cleaning solutions above can only be used for general cleaning. If you use them to control infections, we shall assume no responsibility for the effectiveness.**
-

FOR YOUR NOTES

8 Troubleshooting

8.1 The displaying text on the CMS screen is abnormal.

Please ensure the region and language of the operating system is the same with the language set in the CMS system.

1. To set up the region and language of the operating system, refer to section **3.13.1 *Setting the Region and Language of the Operating System.***
2. To set up the language of the CMS, Click “System Setup” → “Admin Setup” → “Other” → “Language” and select the desired language.

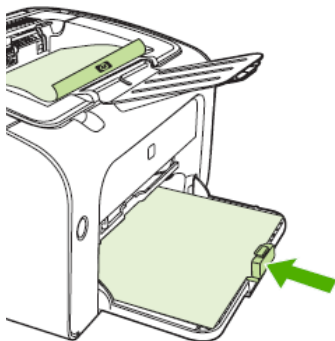
8.2 Recorder/printer-related technical alarm messages

Please refer to the “BeneVision Central Monitoring System Operator’s Manual” Appendix B.

When print fails, please follow the instruction to recover the print job:

1. Restart the printer.
2. By clicking “System Setup” button and then select “Printing Control” sheet.
3. Click the button “Restart Print Service” to resume.
4. Select the printer in “Printing Control” sheet.

Also, after the paper is loaded in the main input tray, push the tab close to the end of paper.



8.3 Network printing fails

Follow the steps below:

1. Click “System Setup” button and then select “Printing Control” sheet. Restart the print job. Try other printers in the printer list. If the problem still exists, turn to the next step.
2. Check whether network printing works in Windows system. If it works, restart the computer, enter the CMS and repeat step 1. If the problem still exists, turn to the next step.
3. Verify the network cable between printer and CMS host is normal.
4. Verify the printer’s IP address and CMS host’s IP address are in the same network segment. Please refer to printer’s accompanying documents or ask the vendor’s customer service for the method of getting printer’s IP address. For example, for HP LaserJet 1505n, press the print button on the panel to print the configuration report which has IP address information. For HP LaserJet 2035n, press the print button on the panel for 5-10s to print the configuration report.
5. If the verification is Ok but the problem still exists, try to delete the printer driver and reinstall it by referring to **1.1** .
6. If the problem still exists after driver installation, please contact the printer vendor’s customer service.

8.4 No waveform displayed in waveform review

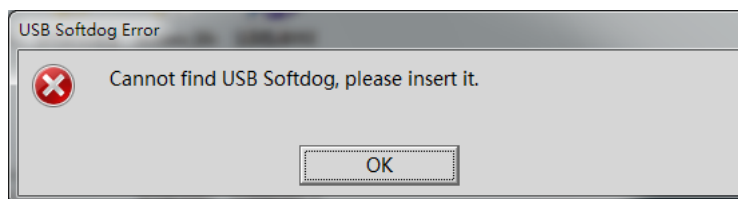
This symptom indicates that the user does not add the waveforms that need to be saved. To remove this fault, click in the patient window and then select “Display Setup” to enter the “Display Setup” tab sheet, select “Wave Save” and then add your desired waveforms.

8.5 Multi-/Dual-screen changes to Single-screen during the CMS installation

1. Check if the displays are connected to host and work functionally.
2. Check if the display mode is set correctly; please refer to **3.7.1 Double Screen Display Adapter**, **3.7.2 Implement Double Screen Display by DP-to-VGA Adapter** or **3.7.6 Installing Multi-Screen Extend Device (SUNIX VGA2715)**.

8.6 Remove Dongle Error

The following prompt message may appear during installation or running.



Dongle error may be caused by dongle driver not installed, dongle inserted improperly, or USB port damaged. Please check dongle driver or reinsert dongle.

8.7 CMS Unable to Connect the Bedside Monitor

This fault may be caused by network configuration, network cable plugged improperly, network cable damaged etc. Refer to the following procedures to troubleshoot this fault.

1. Check if the network cable is plugged properly.
2. Check network configuration to see if the IP address of the CMS and that of network bed number of the monitor are configured correctly.
3. Use ping command to check if the CMS can be connected to the bedside monitor. Try to ping as many monitors as possible. In the case of ping failure, the fault may lie in hardware.
4. Check if the crystal connector on the network cable or the network cable is damaged by using special tools.

8.8 Abnormal Database Service Handling

If database service stops by accident or runs abnormally, it will make the CMS unable to normally save patient data. When this situation occurs, please try to restart the CMS. If the high level system alarm disappears, it indicates that database service restores to normal storage status. If the high level system alarm still persists, it indicates that database is severely damaged. In this case, you need to re-install database software. Please refer to **3.9 Installing Database Software** for detailed operation.

8.9 Database Backup and Recovery in Case of System Failure

You can use the database backup and recovery tool to back up the current database and restore the database. For example, you can back up the current database before you change the hard disk, reinstall the operating system or recover the CMS using the “quick recovery disk”; and then restore the data using the backup database after the operating system has

been installed or the CMS has been recovered.

For database backup and recovery, refer to the operating manual.

8.10 Dual Hard Disks Failure

When the dual hard disks work normally, the dual hard disk icon  displays at the lower right corner of Windows desktop. If the dual hard disks storage system fails, the following icon will show:

-  Indicates that the storage system is in a warning state and data may be at risk.
-  Indicates that the storage system is in an error state and data may be lost.

If these icons display, immediately contact the local HP service.

When the icon  displays, it indicates that the system is rebuilding. Please wait until the operation finishes.

NOTE

-
- **Before replacing the fault disk, you should exit the CMS and enter the Windows screen. You can select “Factory Setup” → input the correct password → “Exit to Windows ” to enter the Windows desktop.**
-

8.11 eGateway Unable to Obtain the Exported Files from the CMS

If the CMS exported files cannot be received on the eGateway, you need change the settings on CMS and eGateway as followings.

Change the Firewall Settings on the CMS Host

1. Open “Control Panel”.
2. Select “Windows Firewall”, and then “Allow a program or feature through Windows Firewall”.
3. Allow “File and Printer Sharing” and “netlogon service” applications to pass through all network firewalls.

Change the SMB Authentication Level on the eGateway Host

1. Run "GPedit.msc".
2. Select "Computer Configuration" → "Windows Settings" → "Security Settings" → "Local Policies" → "Security Options".
3. Double click "Network security: LAN Manager authentication level". The properties screen displays.
4. Set "Network security: LAN Manager authentication level" value to "Send LM & NTLM – use NTLMv2 session if negotiated".
5. Run "Services.msc", and restart the "Server" service.

FOR YOUR NOTES

